

Based on operation the operators divided into several types.
They are

1. **Assignment operator [=]:** It copies the value on its right side in to the variable on its left side. In assignment operation the left side operand should be a variable i.e. left side constant or expressions not allowed.

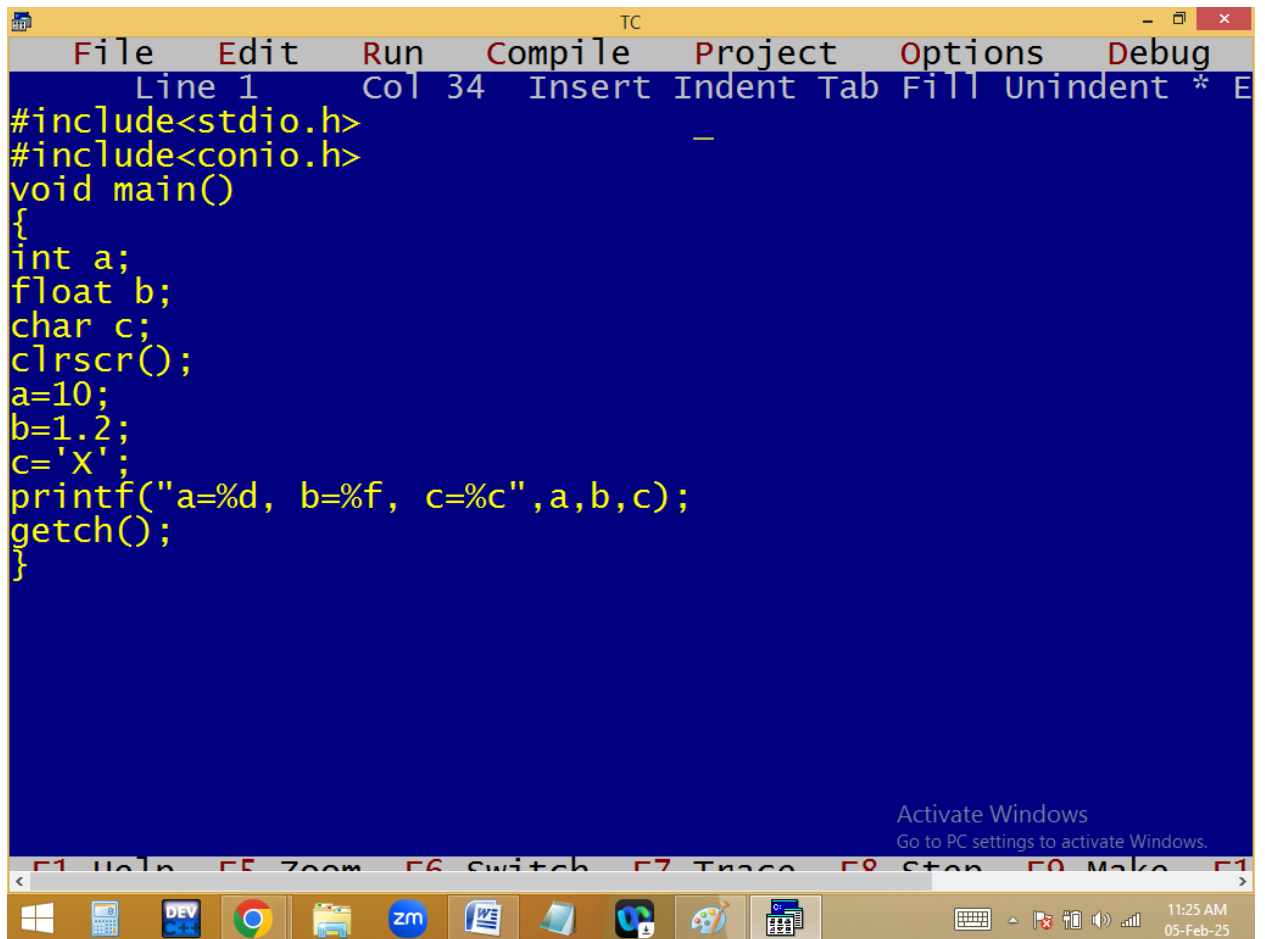
Eg:

a=10;

b=1.2;

c='X';

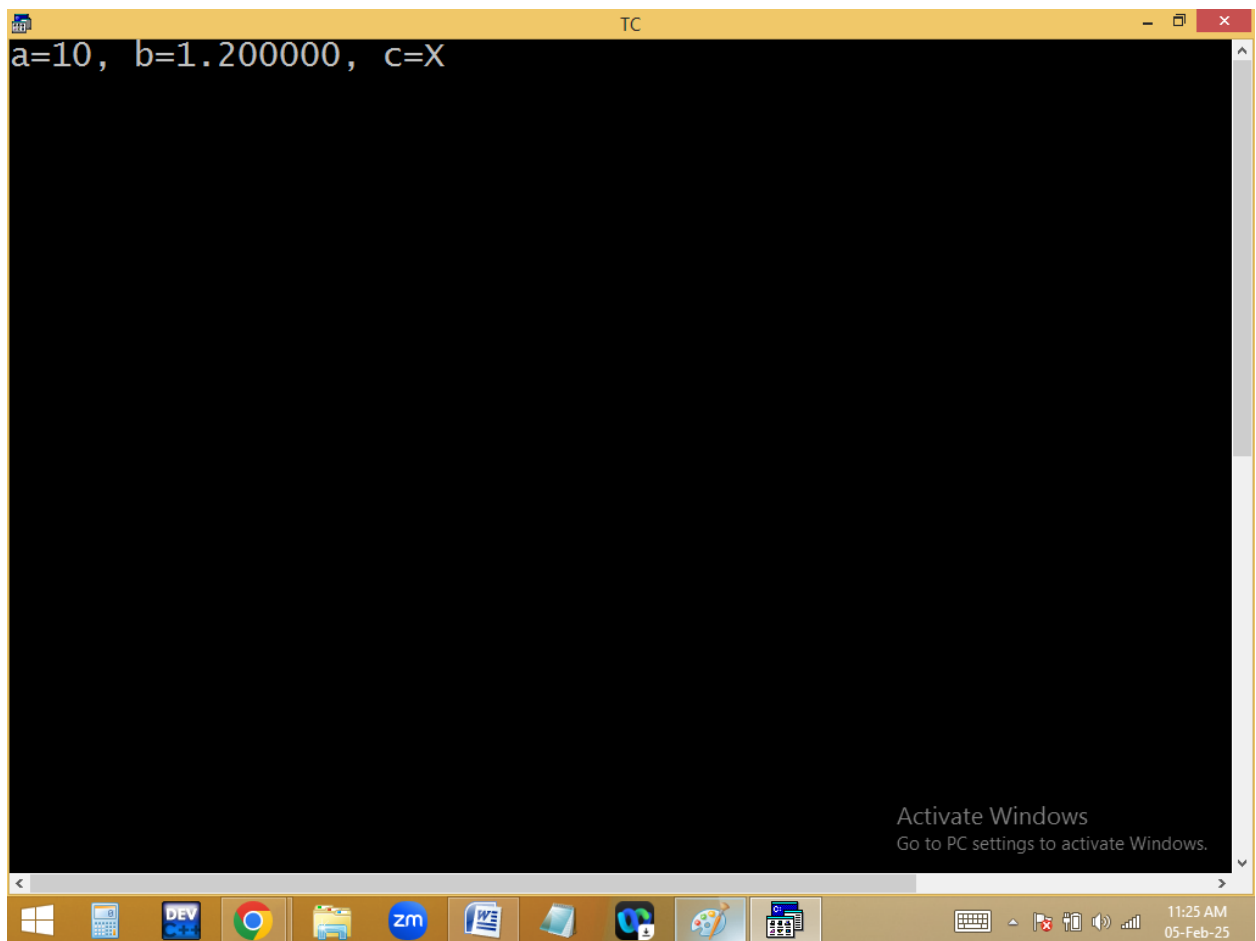
d="abcd"; ➔ Error because of abcd is a string

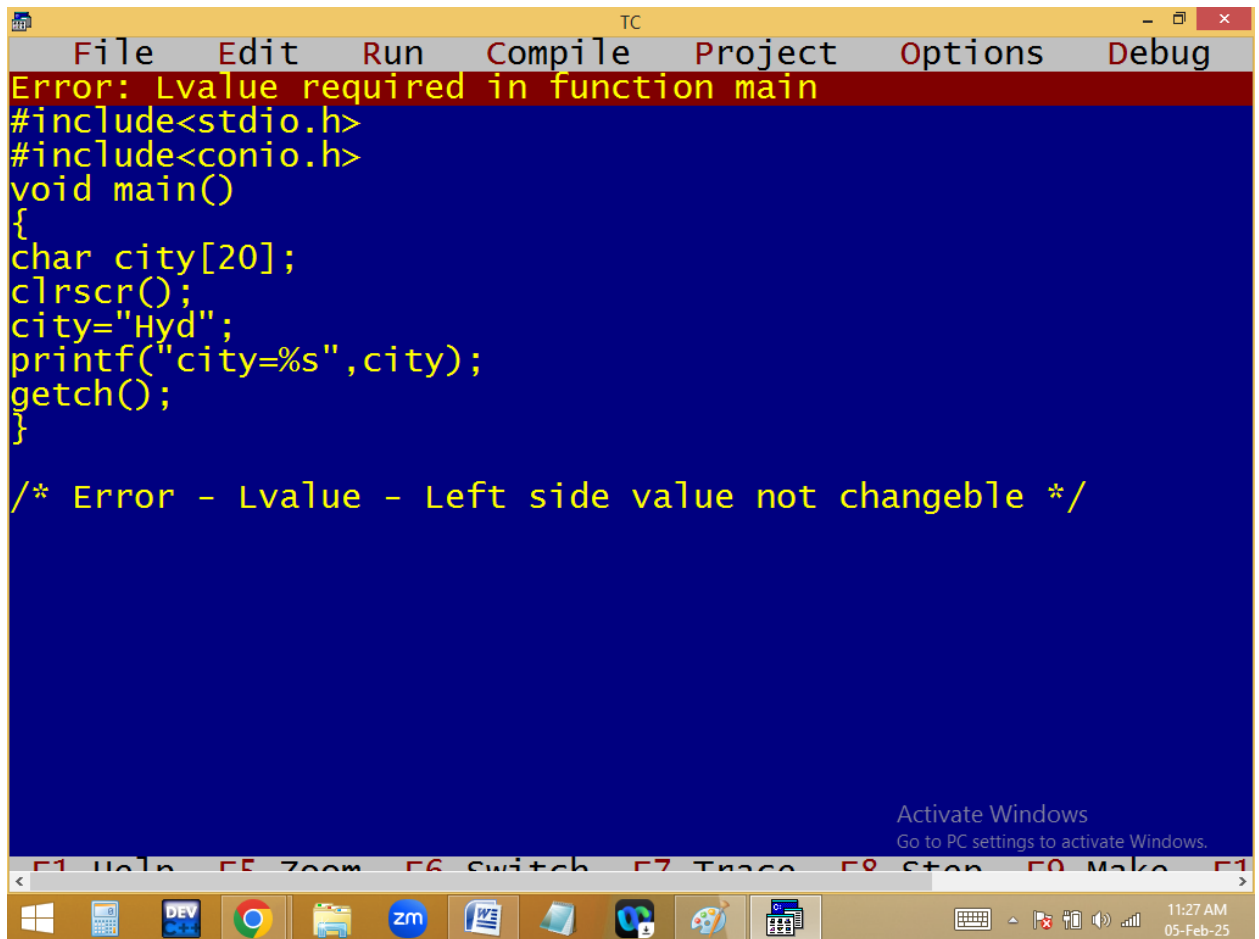


```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
a=10;
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);
getch();
}

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

Windows taskbar icons: Windows Start, Task View, DEV, Chrome, File Explorer, Zoom, Word, Edge, Paint, Calculator, System tray (Keyboard, Network, Volume, Battery, Date/Time: 11:25 AM 05-Feb-25).





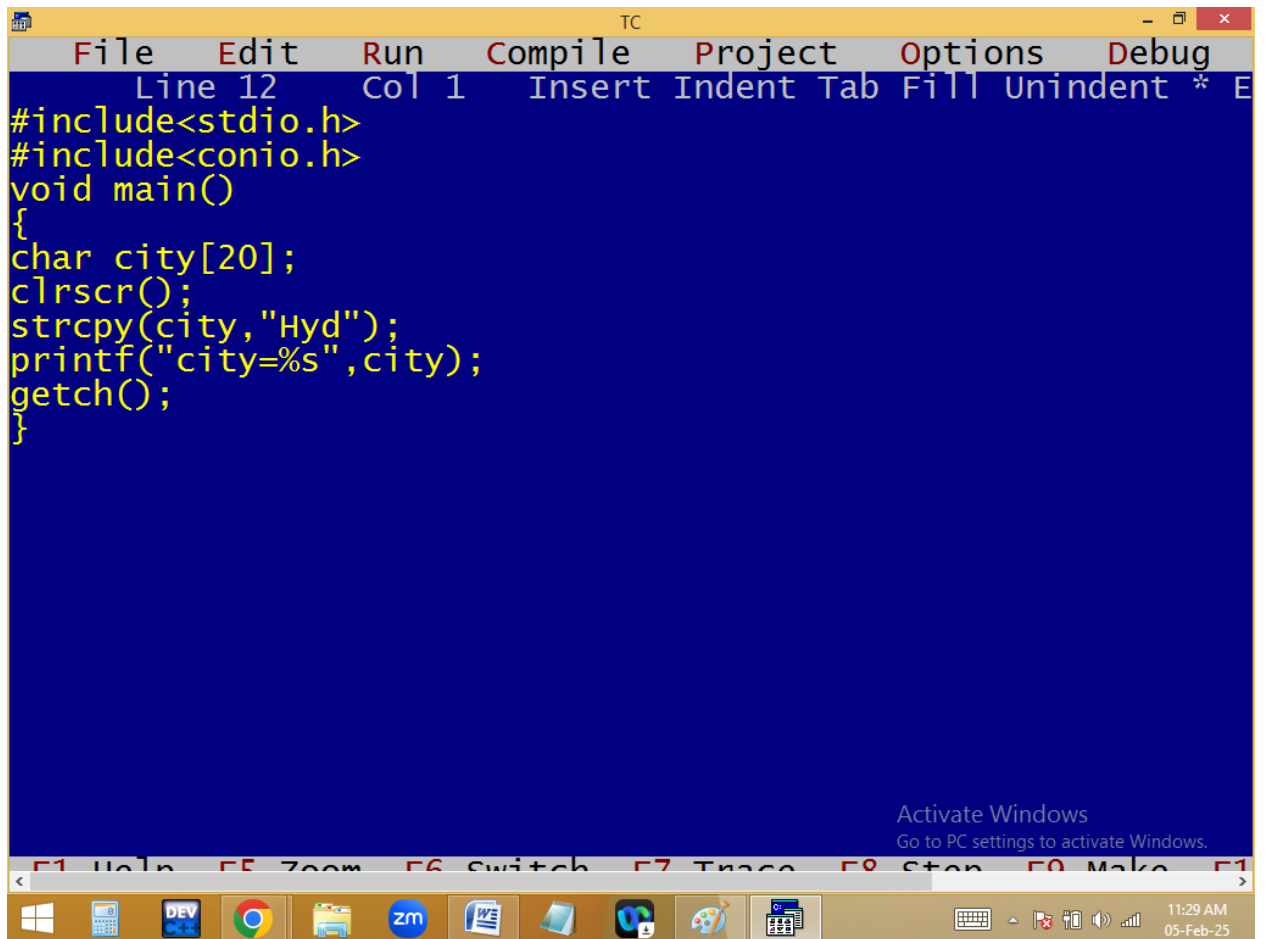
The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. A red error banner at the top reads "Error: Lvalue required in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char city[20];
clrscr();
city="Hyd";
printf("city=%s",city);
getch();
}

/* Error - Lvalue - Left side value not changeble */
```

At the bottom of the IDE, there is an "Activate Windows" watermark and a row of function key shortcuts (F1-F11). The Windows taskbar at the very bottom shows various application icons and the system clock indicating 11:27 AM on 05-Feb-25.

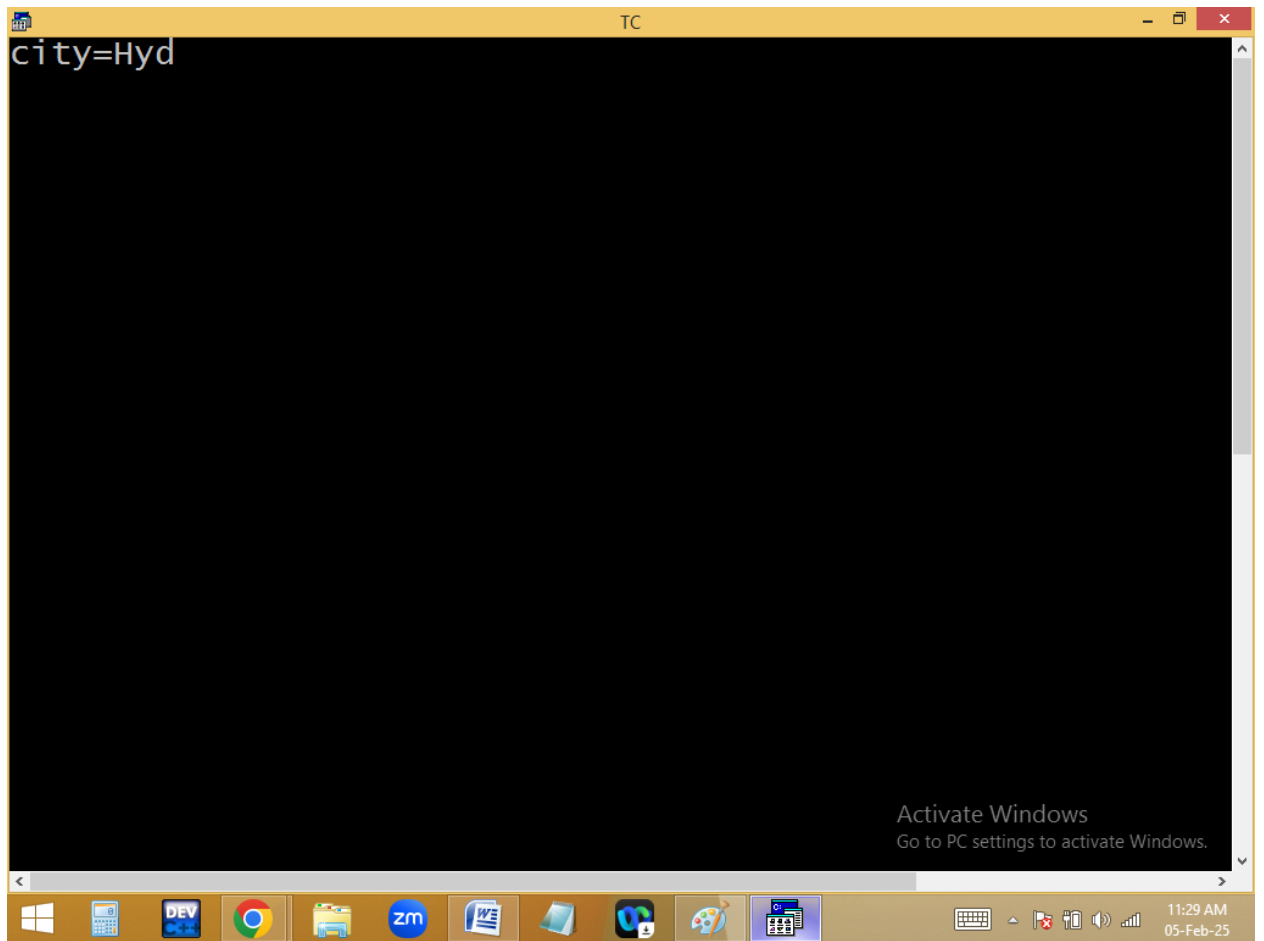
Note: To copy a string we have to use strcpy() function available in <string.h>

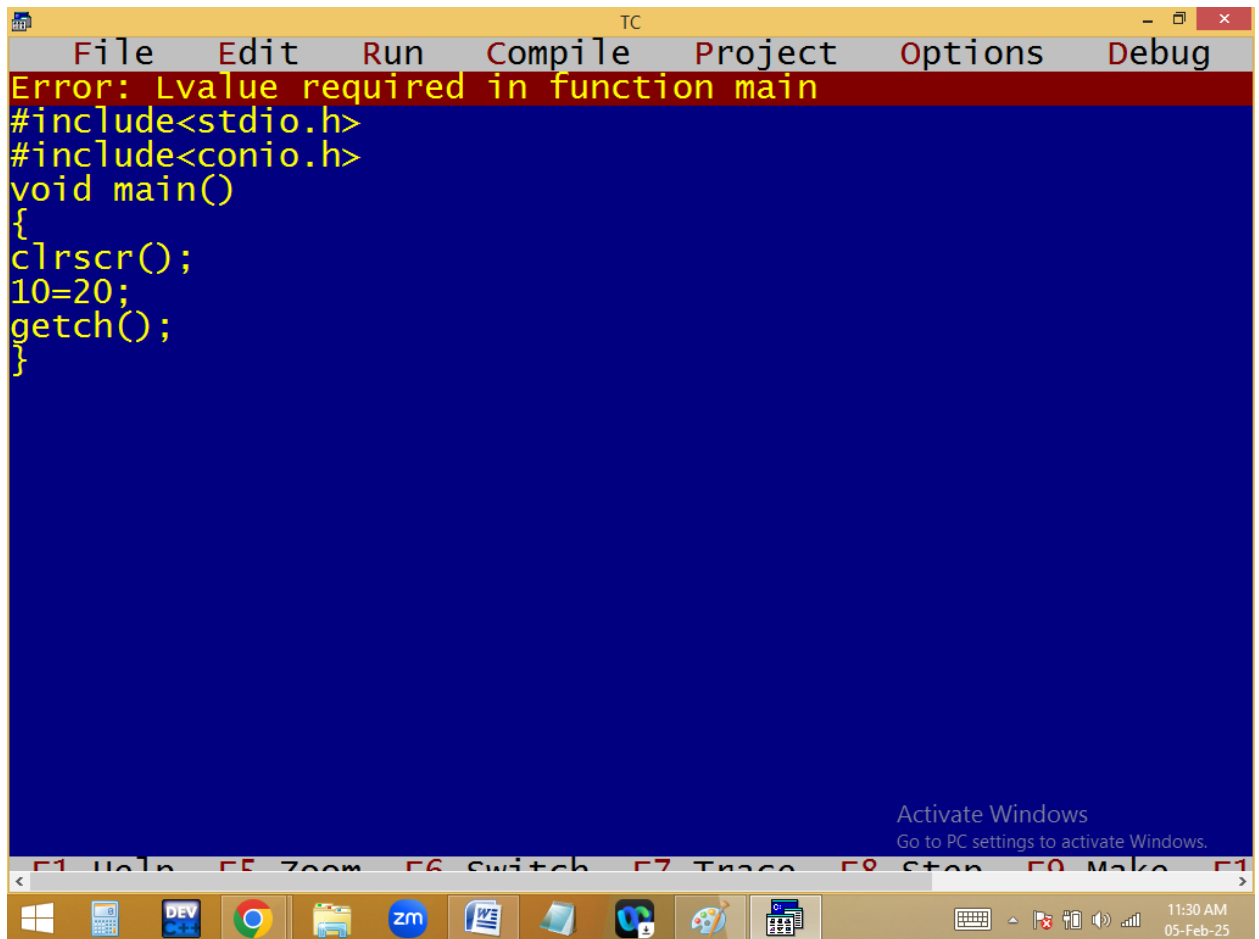


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar displays "Line 12", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char city[20];
clrscr();
strcpy(city,"Hyd");
printf("city=%s",city);
getch();
}
```

At the bottom of the window, there is a Windows taskbar. It contains several application icons: Windows Start button, Task View, File Explorer, Google Chrome, a folder icon, a blue circular icon with "zm", a document icon with "WE", a folder icon, a blue circular icon with a white "a", a paint palette icon, and a calculator icon. The system tray on the right shows the date and time: "11:29 AM" and "05-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

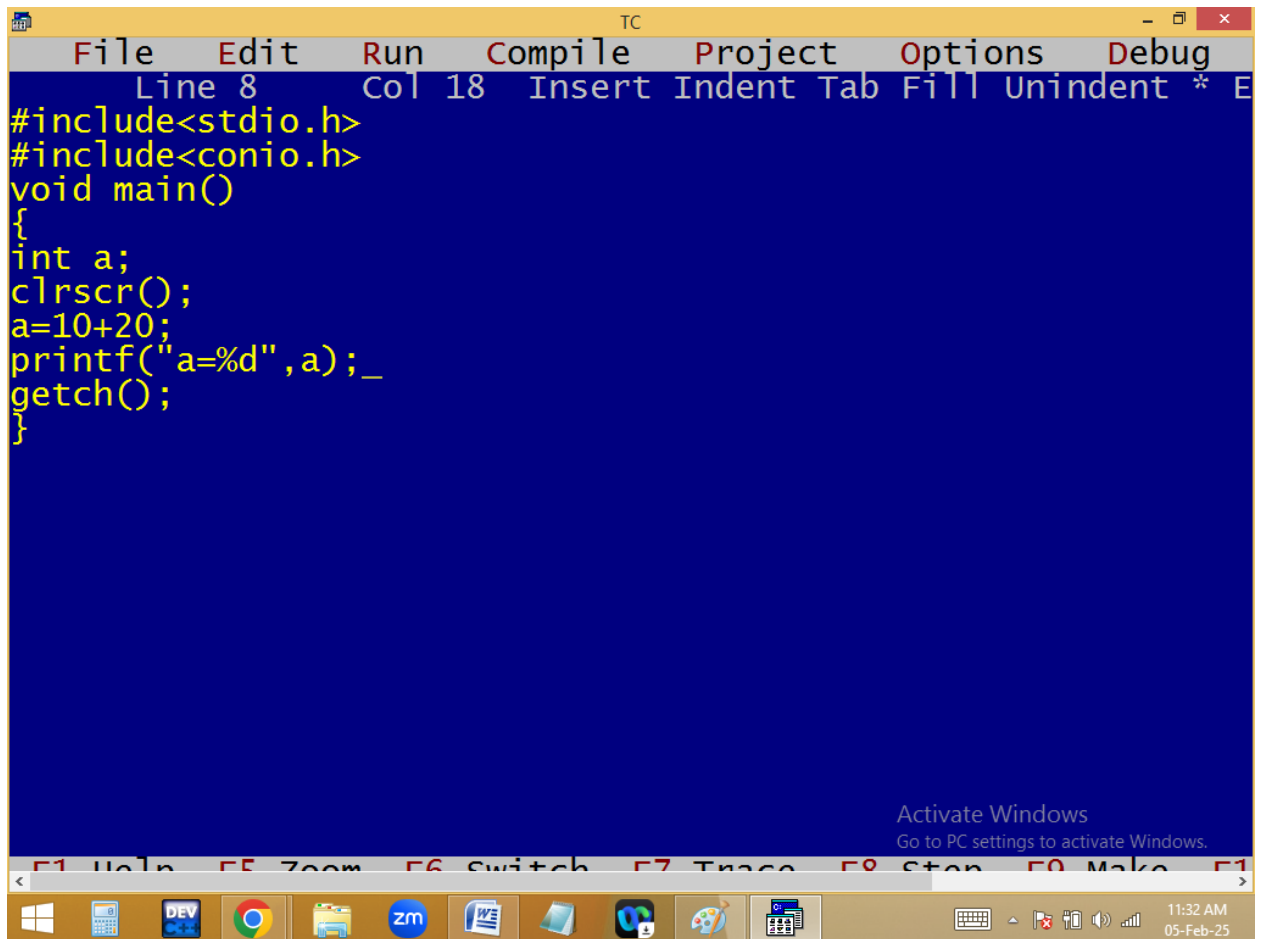




The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor area states "Error: Lvalue required in function main". The code in the editor is as follows:

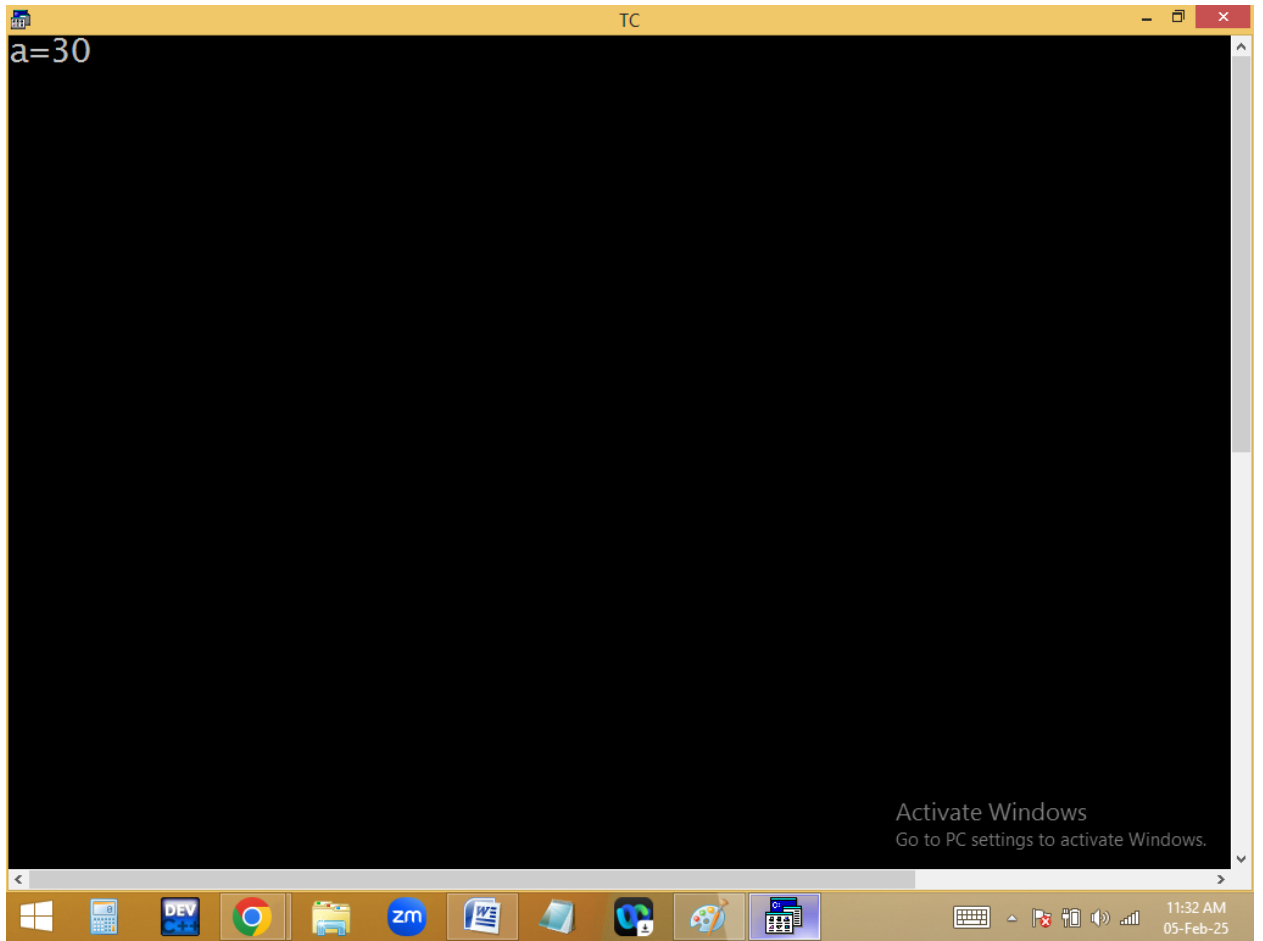
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
10=20;
getch();
}
```

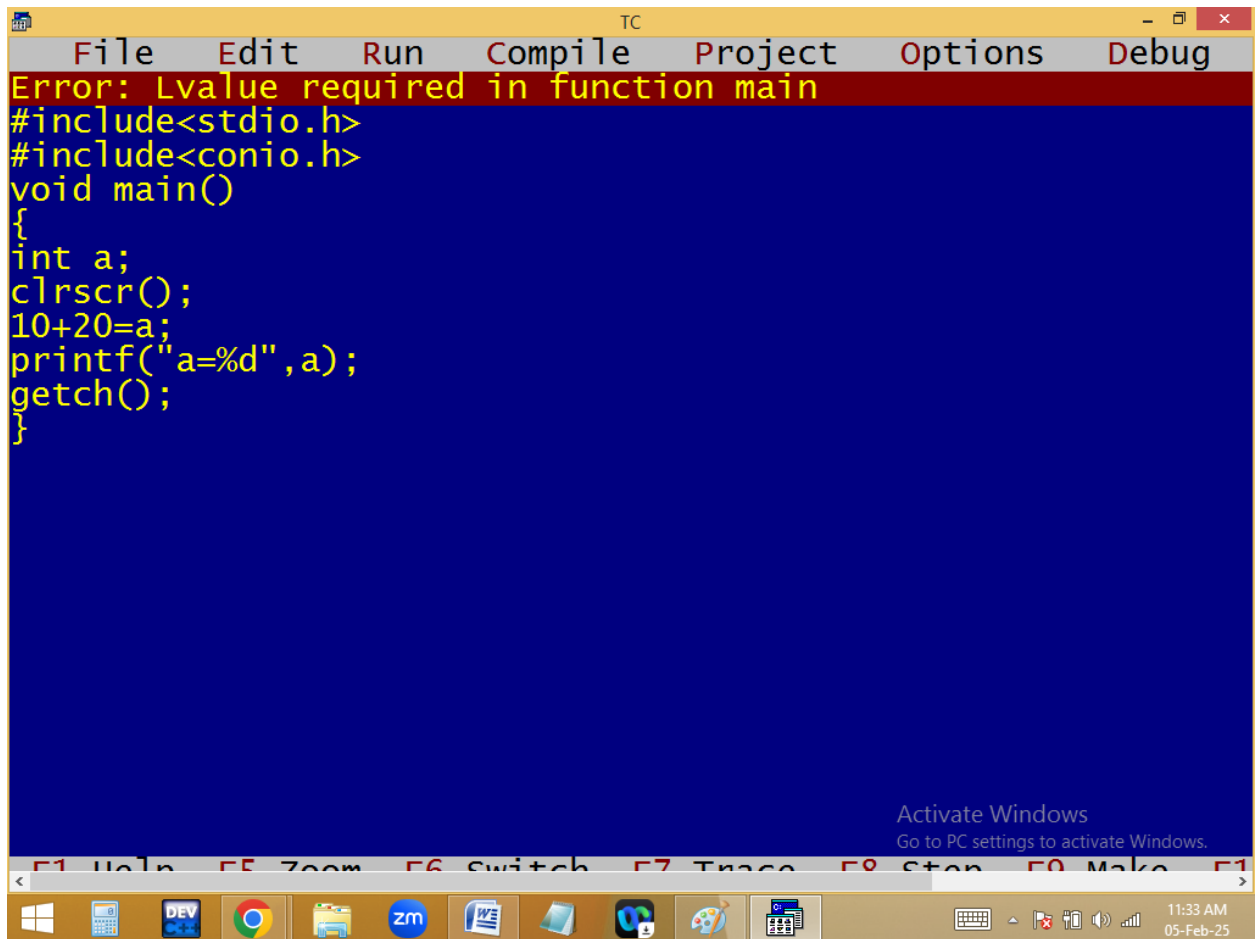
At the bottom of the window, there is a taskbar with various application icons including Windows, Calculator, DEV C++, Chrome, File Explorer, Zoom, Word, and others. The system tray on the right shows the time as 11:30 AM and the date as 05-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 18 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a=10+20;
printf("a=%d",a);_
getch();
}

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



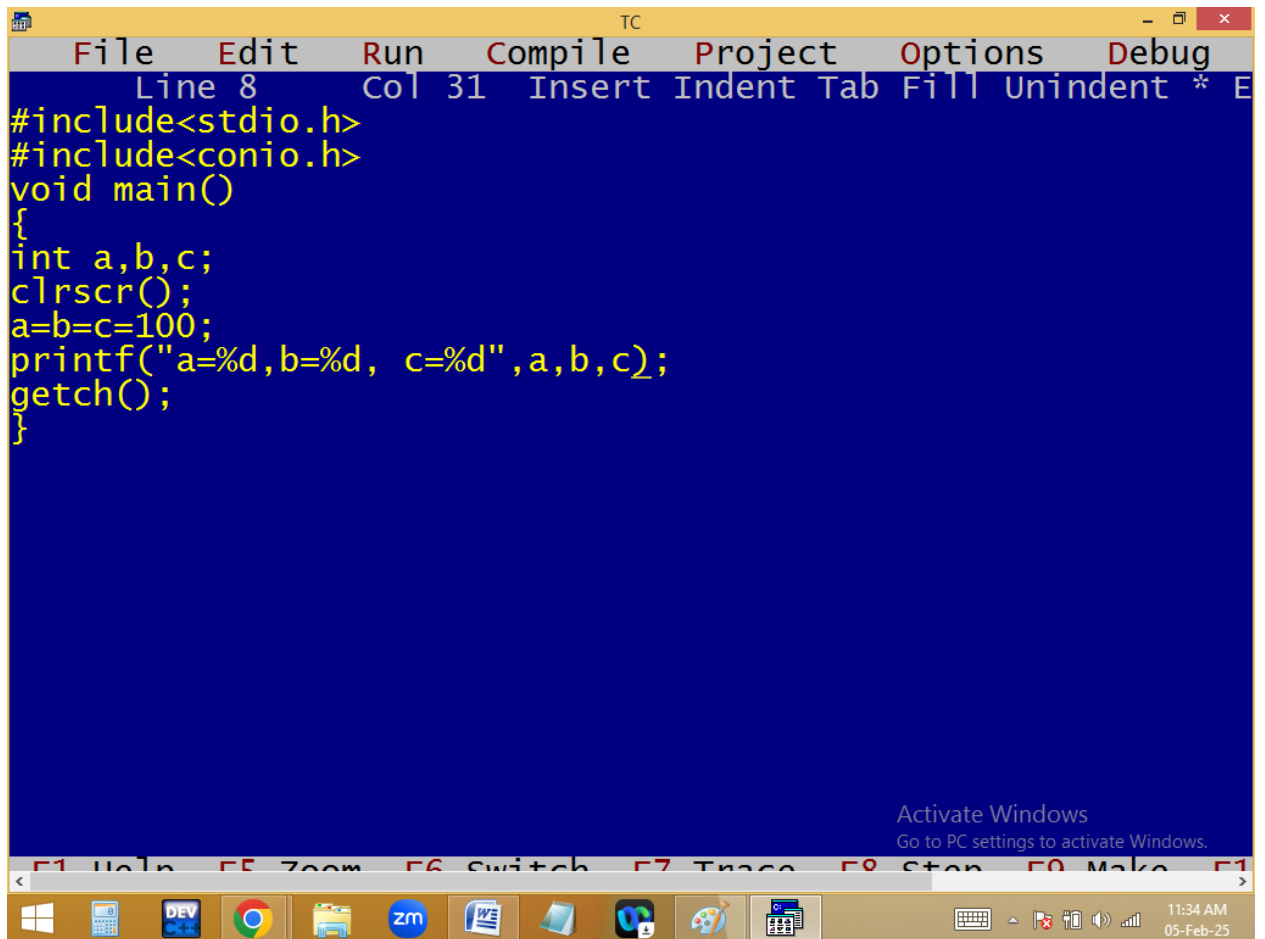
The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. A red error banner at the top reads "Error: Lvalue required in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
10+20=a;
printf("a=%d",a);
getch();
}
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

Below the IDE window, the Windows taskbar is visible, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows 11:33 AM on 05-Feb-25.

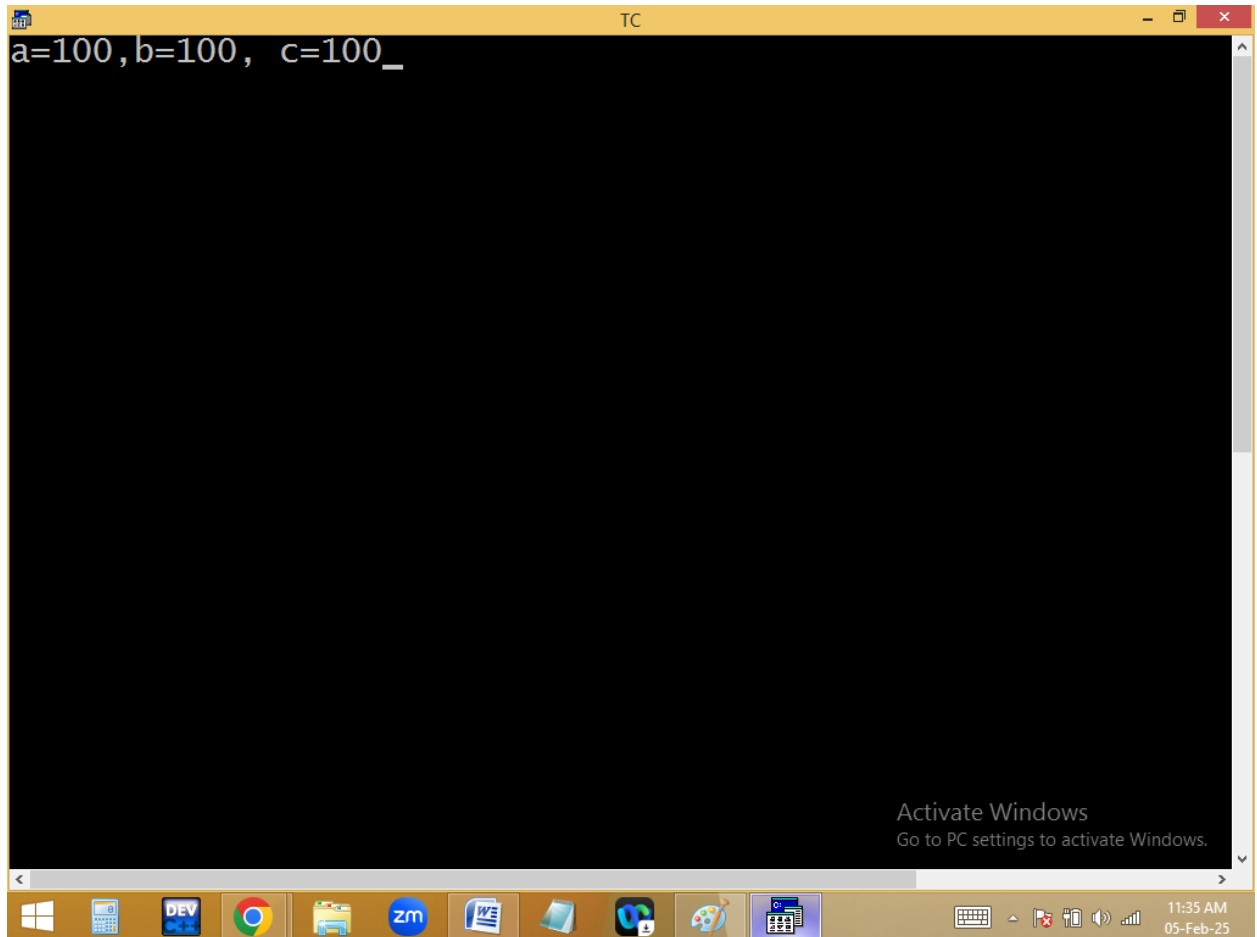
a=b=c=100;



The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a blue background with yellow text. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 8' and 'Col 31'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
a=b=c=100;
printf("a=%d,b=%d, c=%d",a,b,c);
getch();
}
```

At the bottom of the IDE window, there is a text box that says 'Activate Windows' and 'Go to PC settings to activate Windows.' Below the IDE window, the Windows taskbar is visible, showing icons for Windows, File Explorer, DEV, Chrome, File Explorer, Zoom, Word, and several other applications. The system clock in the bottom right corner shows '11:34 AM' and '05-Feb-25'.



2. Arithmetic operators [+, -, *, %, /]: They are used to do the mathematical calculations.

Eg: $a+b$, $a-b$, $a*b$,...

Modules [%] - Remainder:

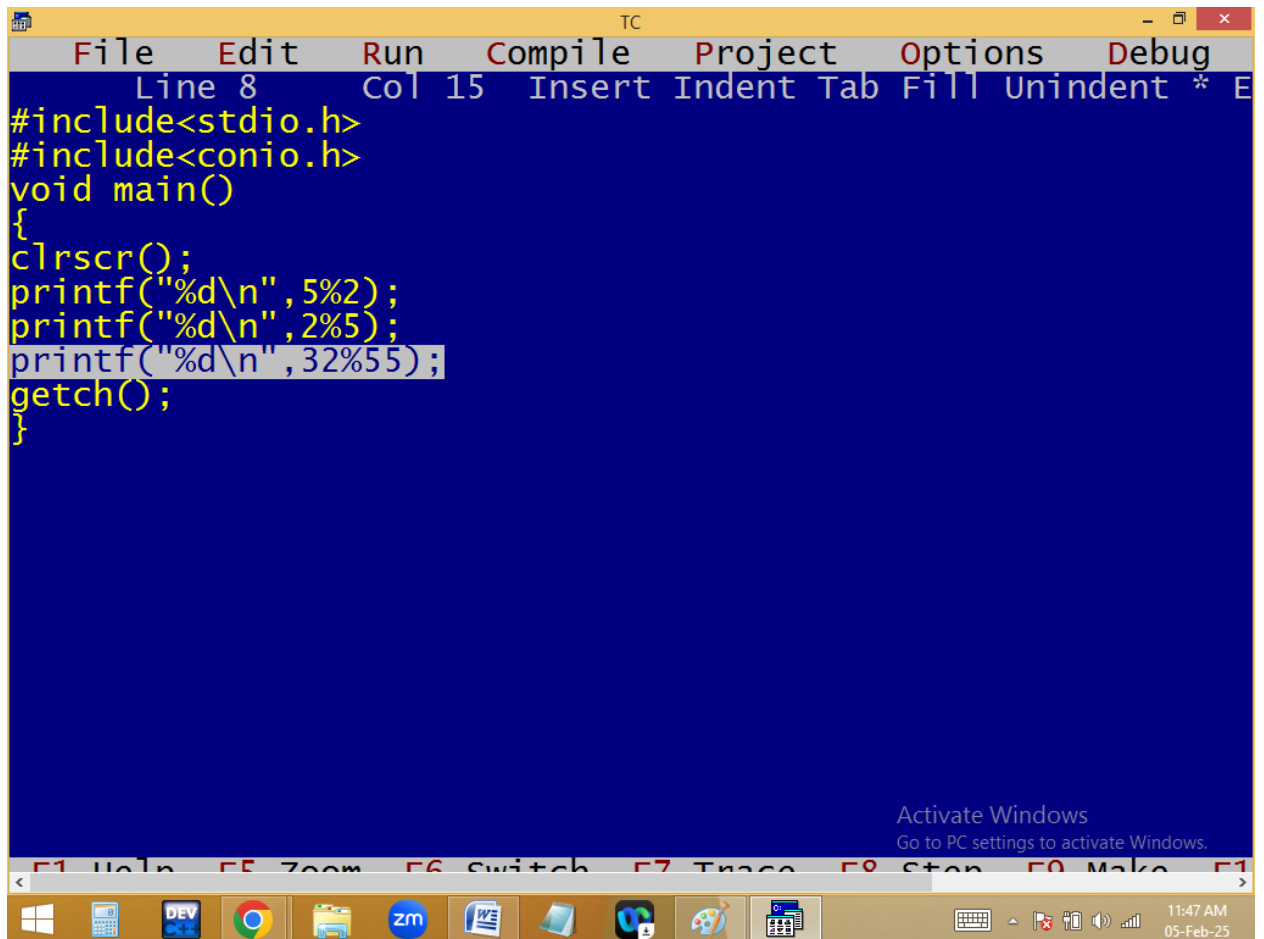
$5\%2=1$

$$\begin{array}{r}
 2 \overline{) 5} \\
 \underline{4} \\
 1
 \end{array}
 \left(2 \leftarrow \text{Quotient } \leq / \right.$$

$\Rightarrow \% \Rightarrow \} \text{ Remainder}$

$$2\%5=2$$

Note: if the divisor bigger than dividend then dividend is the answer.



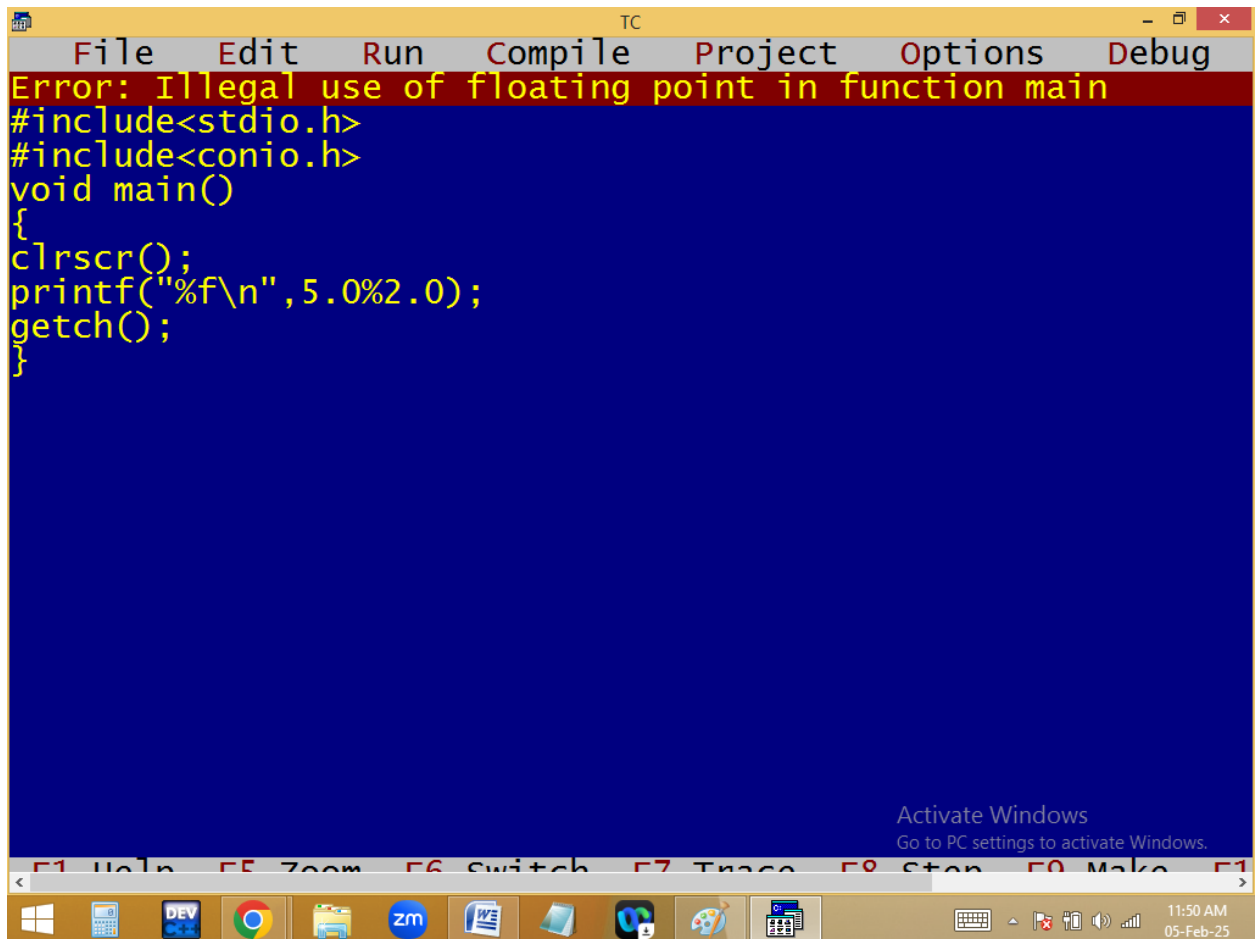
The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates Line 8, Col 15. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5%2);
printf("%d\n",2%5);
printf("%d\n",32%55);
getch();
}
```

The line `printf("%d\n",32%55);` is highlighted. At the bottom of the IDE window, a message reads: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Zoom, Word, and other applications. The system clock shows 11:47 AM on 05-Feb-25.

5.0%2.0=Error

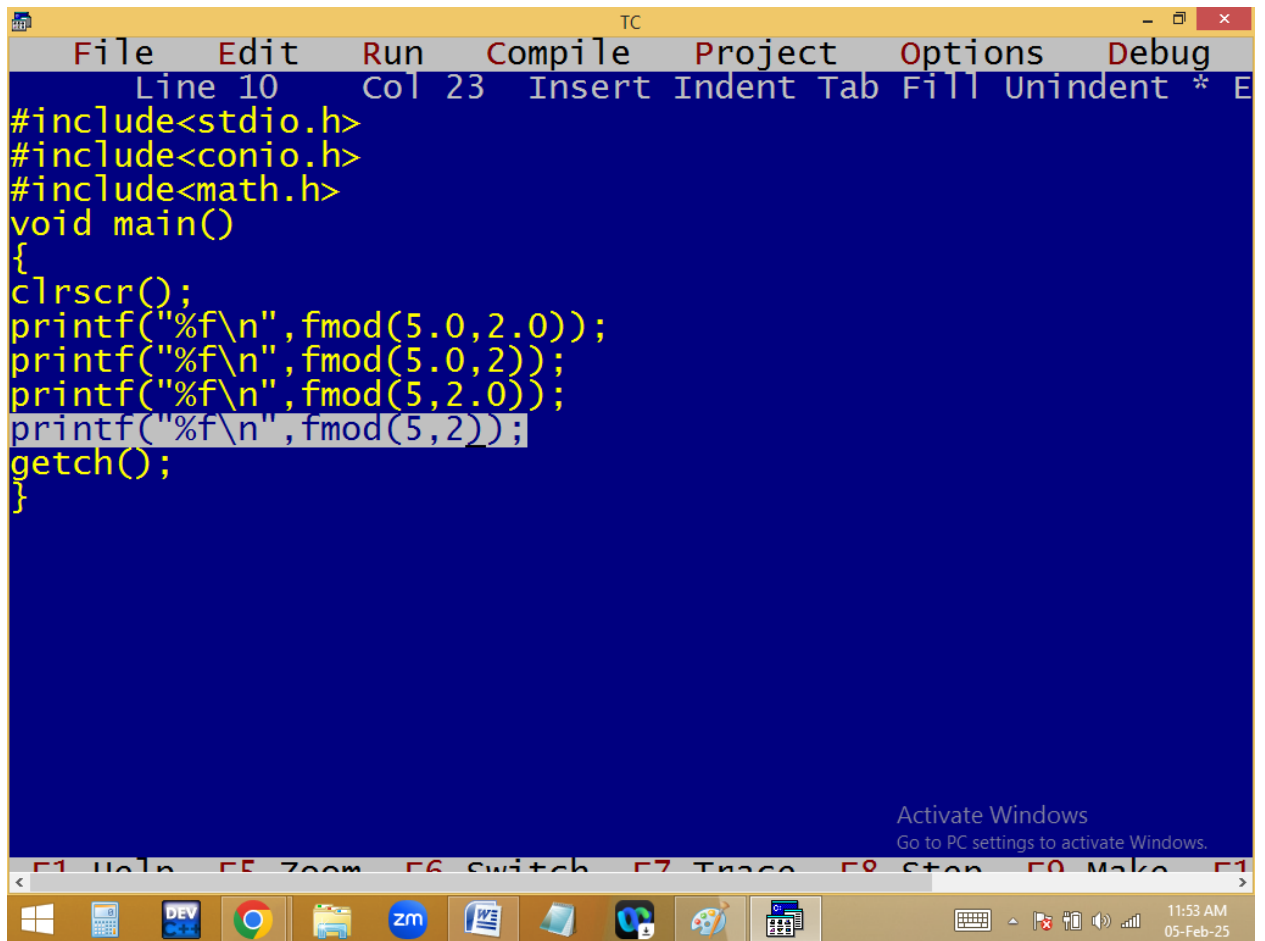
Note: In C & C++ we can't do floating modulus with % operator. We have to use `fmod()` for floating modulus.



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the editor area states: "Error: Illegal use of floating point in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5.0%2.0);
getch();
}
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and several other applications. The system tray on the right shows the date and time as 11:50 AM on 05-Feb-25.

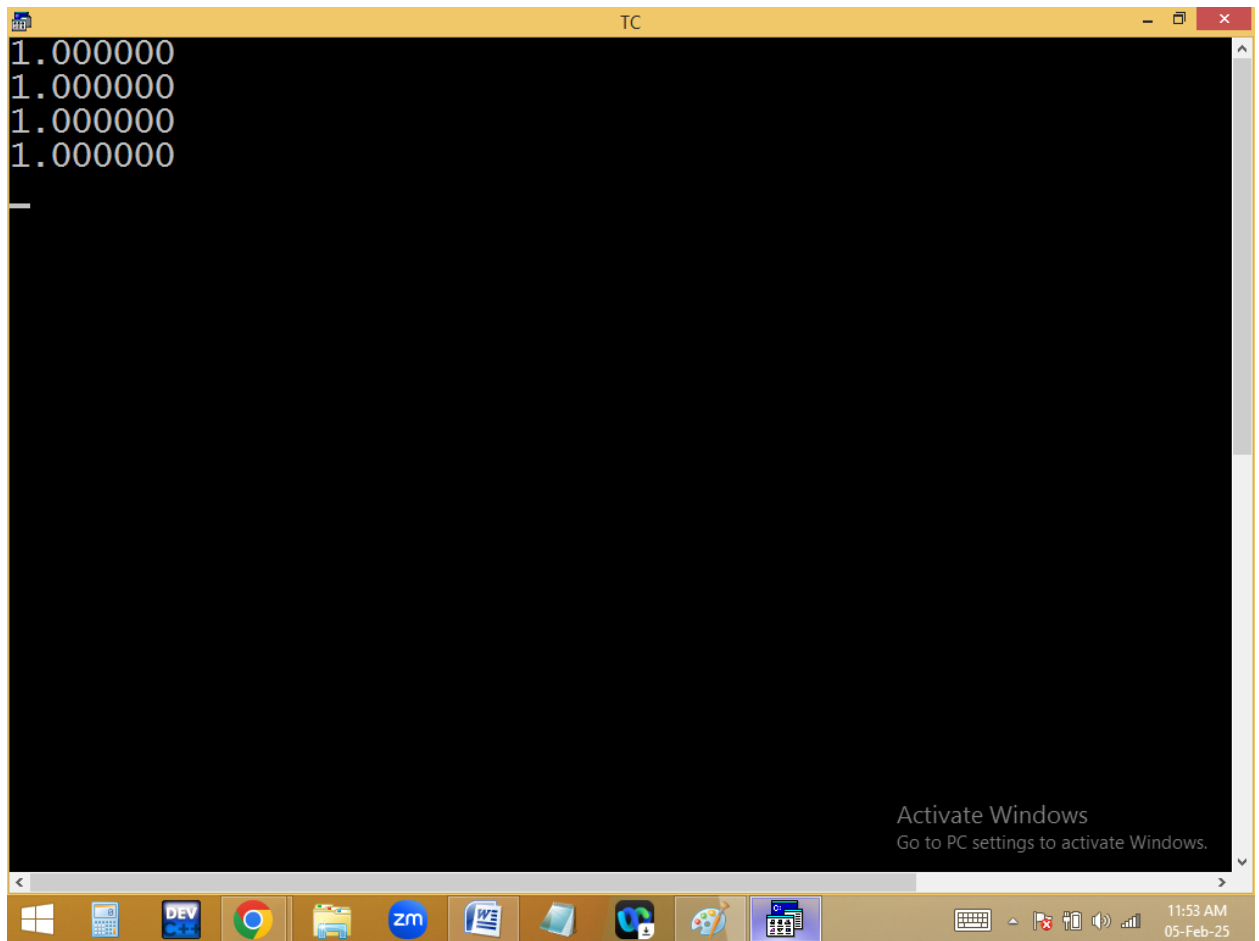


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 23 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5.0,2.0));
printf("%f\n",fmod(5.0,2));
printf("%f\n",fmod(5,2.0));
printf("%f\n",fmod(5,2));
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: Windows logo, Calculator, DEV, Chrome, File Explorer, Zoom, Word, Edge, Paint, Task Manager, System tray (Keyboard, Network, Volume, Battery, Date/Time: 11:53 AM 05-Feb-25)

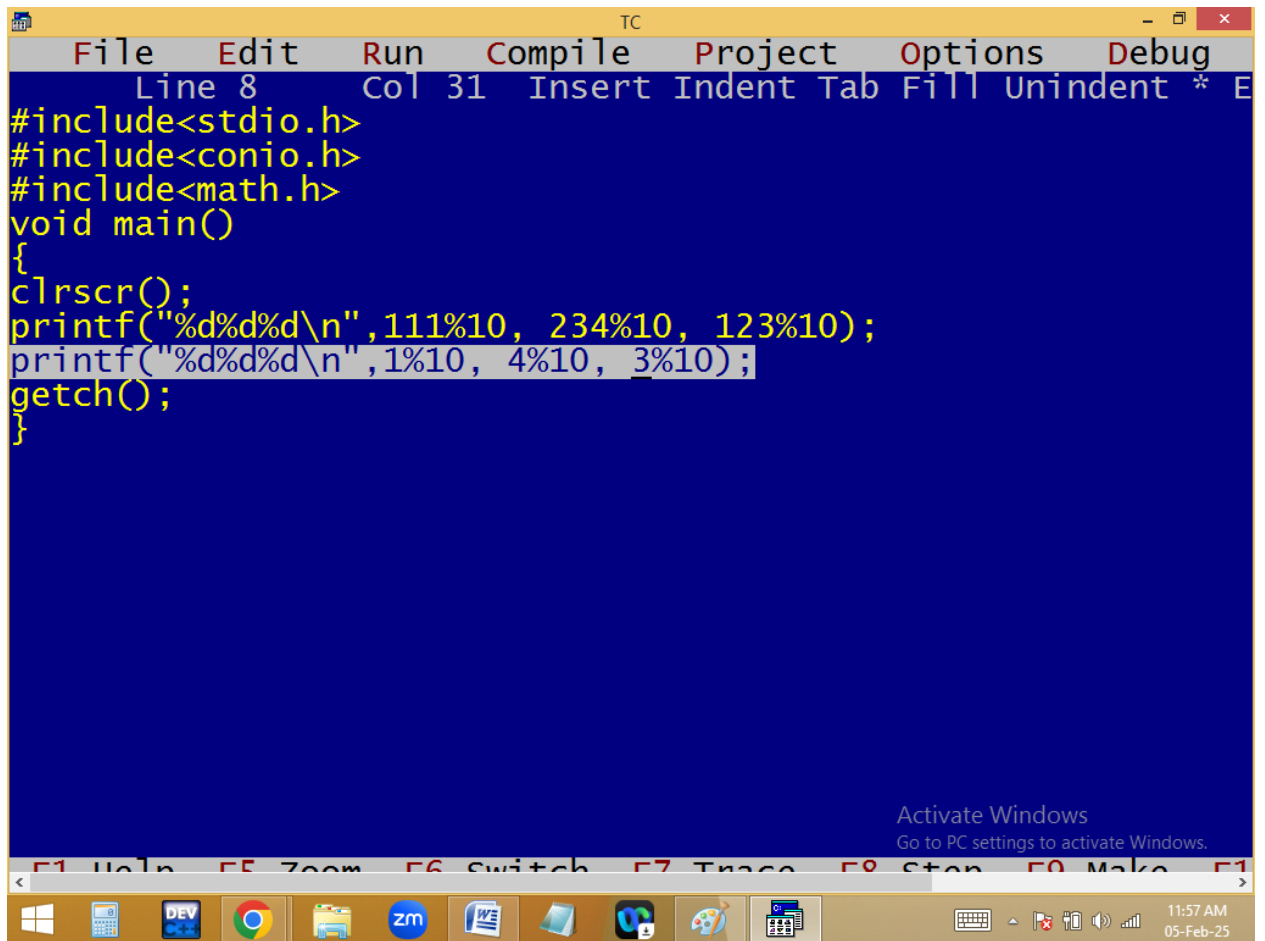


$$111\%10=1$$

$$124\%10=4$$

$$273\%10=3$$

Note: Any $no\%10$ gives last digit.



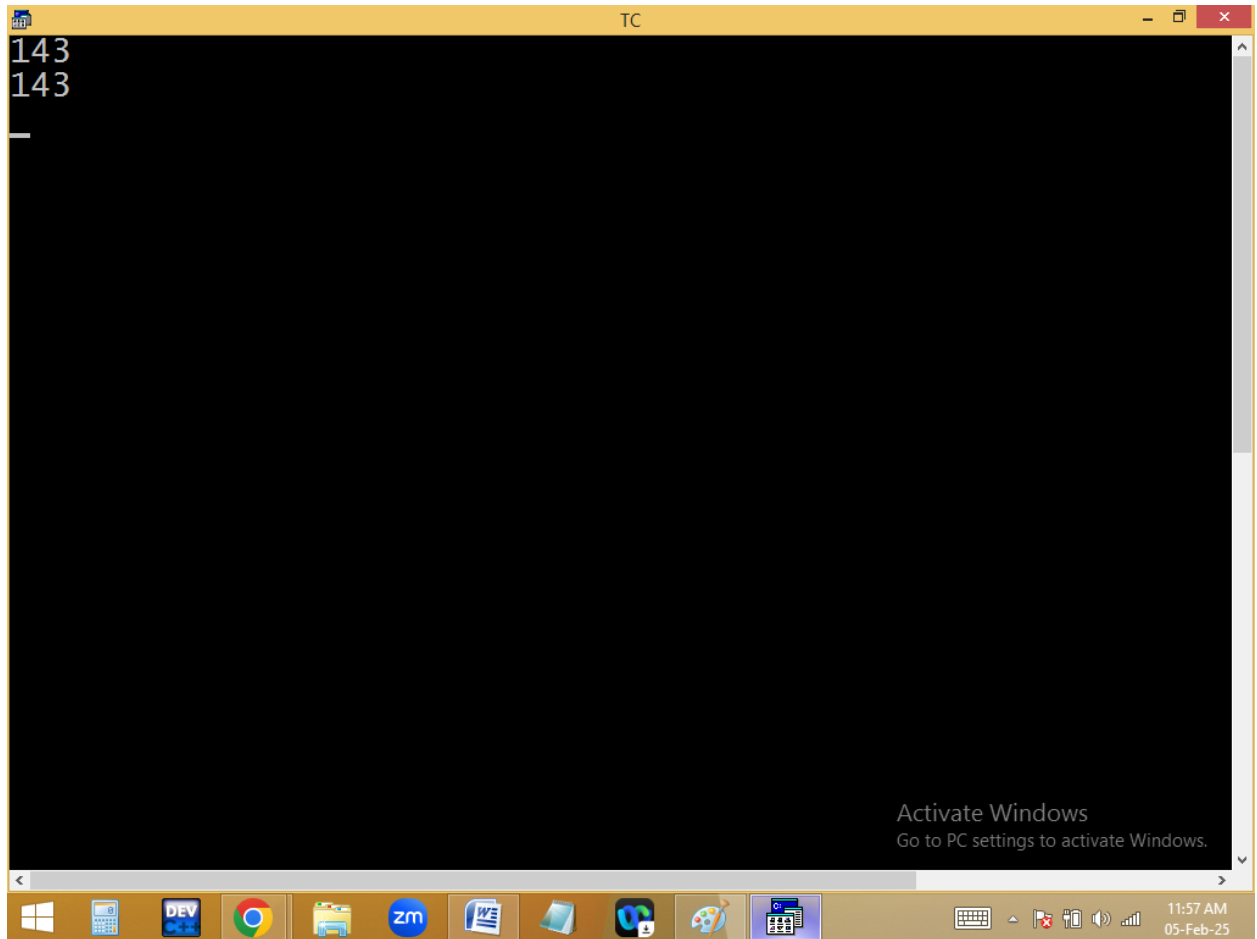
```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 31 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%d%d%d\n",111%10, 234%10, 123%10);
printf("%d%d%d\n",1%10, 4%10, 3%10);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: Windows, Calculator, DEV, Chrome, File Explorer, Zoom, Word, Edge, Paint, Task Manager.

System tray: 11:57 AM, 05-Feb-25.

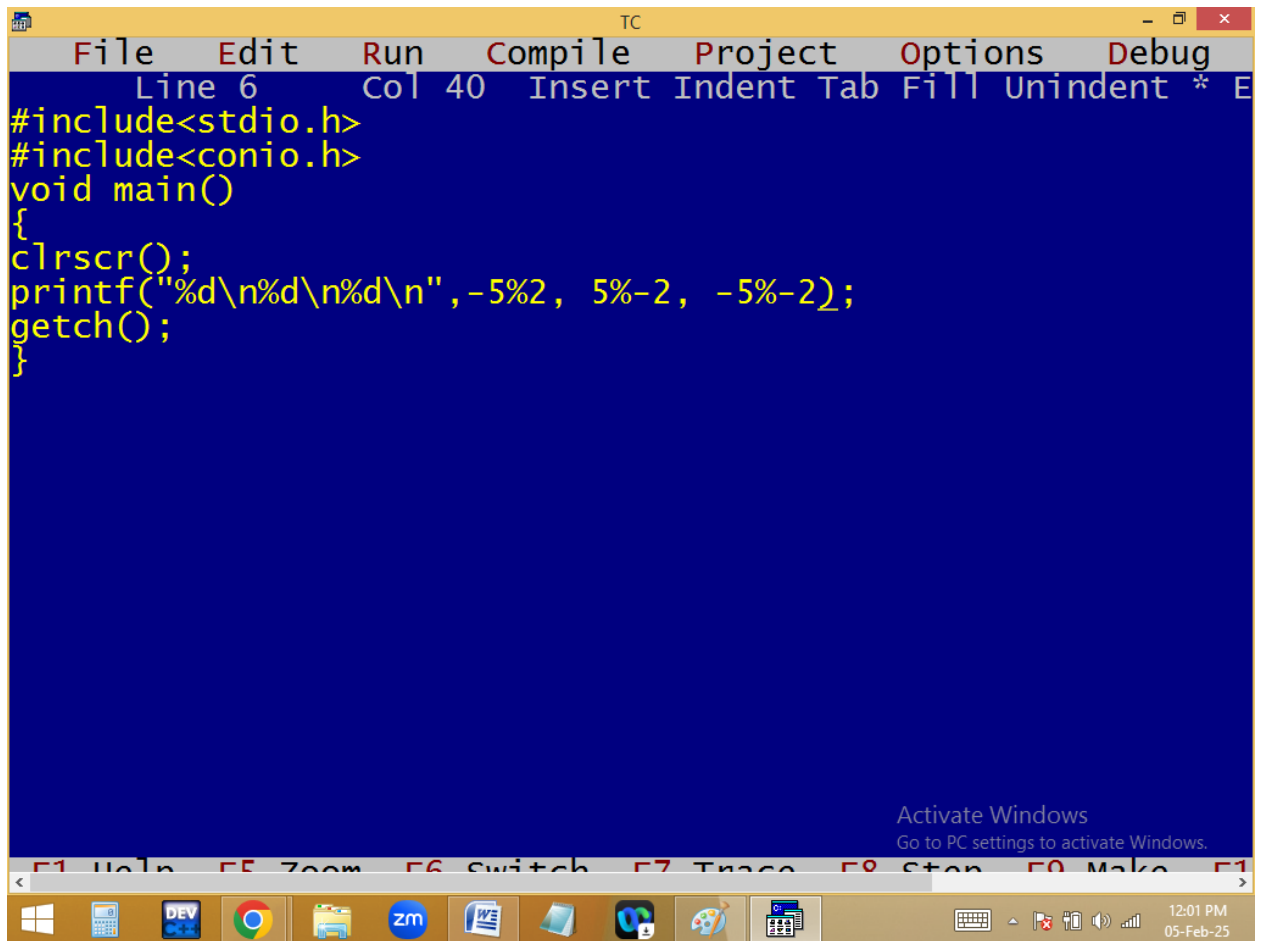


$$\text{-5\%2= -1}$$

$$5\%-2= 1$$

$$\text{-5\%-2= -1}$$

Note: In modules if the numerator is negative then answer also negative.



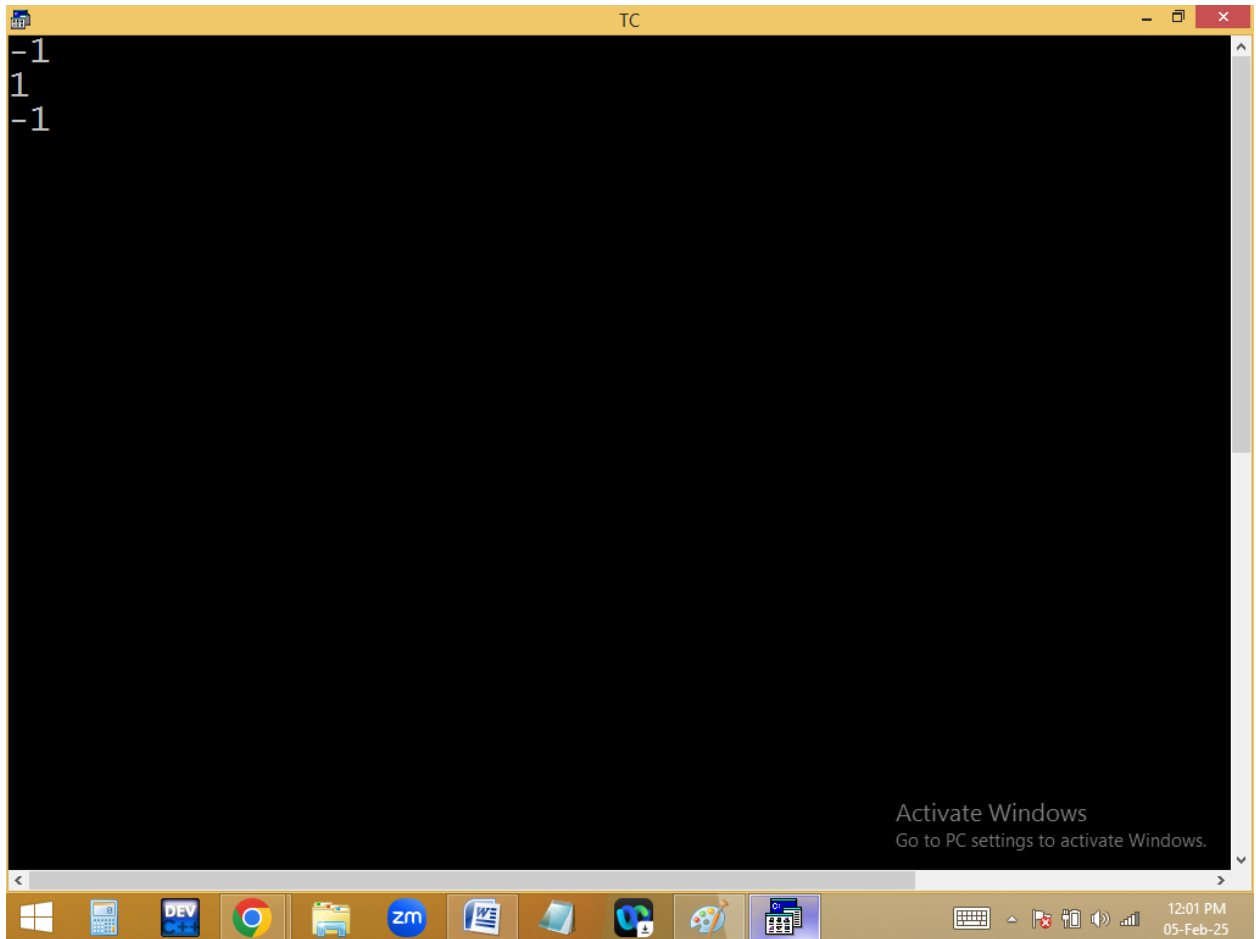
```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 40 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d\n",-5%2, 5%-2, -5%-2);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: Windows, Calculator, DEV, Chrome, File Explorer, Zoom, Word, Edge, Paint, Task Manager, Keyboard, Mouse, Volume, Network, System Tray.

12:01 PM
05-Feb-25



The screenshot shows a Turbo C++ (TC) compiler window with a black background. In the top-left corner, the following text is displayed:

```
-1
1
-1
```

The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 12:01 PM on 05-Feb-25. An "Activate Windows" watermark is present in the bottom-right corner of the window.

/ - division [Quotient]:

$$5/2=2$$

Note: In C & C++ both operands are int then result also int.
if anyone or both are floats then result also float.

$$5.0/2=2.500000$$

$$5/2.0=2.500000$$

$$5.0/2.0=2.500000$$

$$(\text{float})5/2=2.500000 \text{ [Explicit type casting]}$$

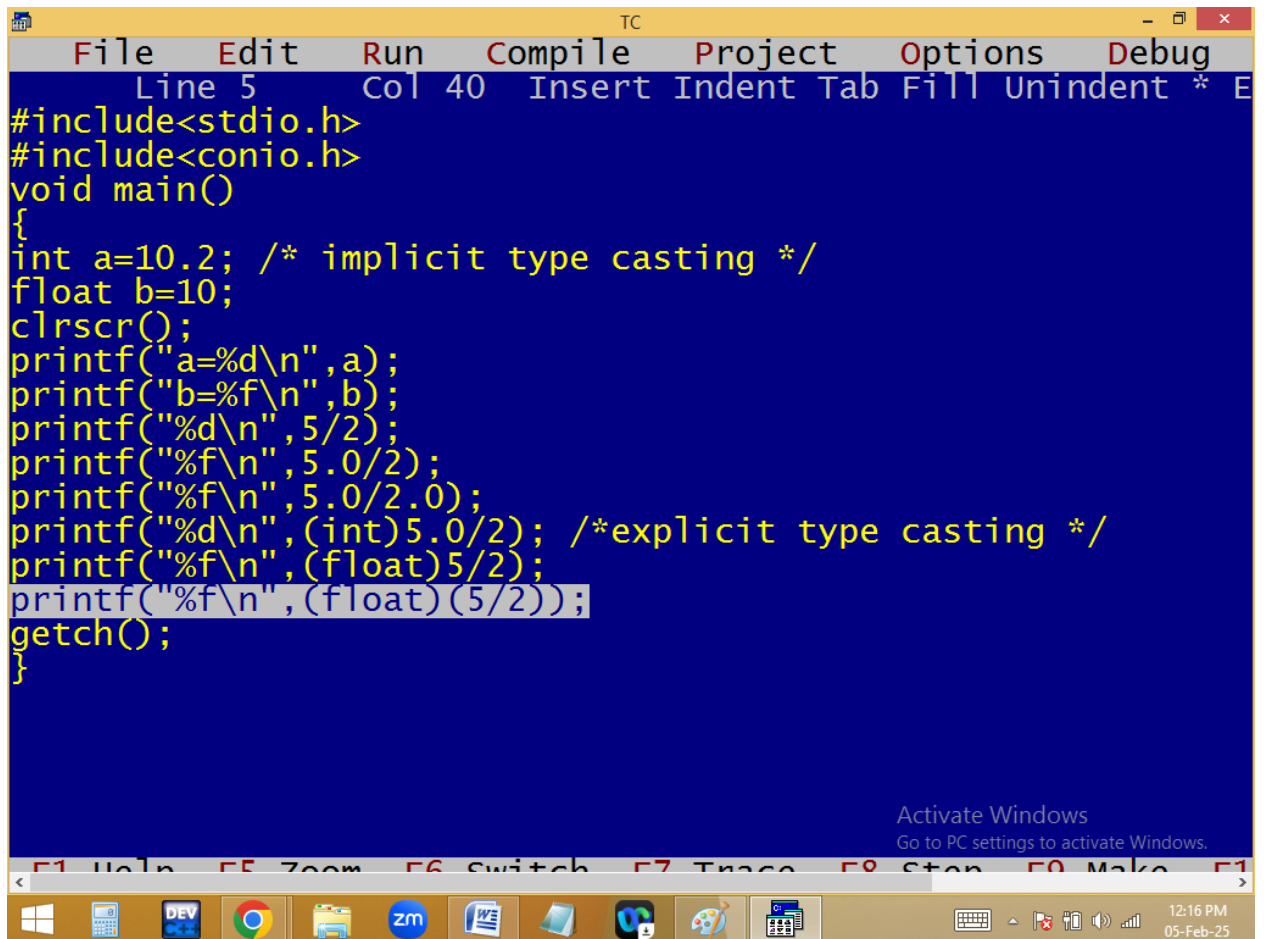
$$5/(\text{float})2=2.500000 \text{ [Explicit type casting]}$$

(int) 5.0/2 = 2 /* explicit type casting */

int a=1.2; ➔ a=1 ← implicit type casting

float b=10; ➔ b = 10.000000 ← implicit type casting

(float) (5/2) = 2.000000



```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 40 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.2; /* implicit type casting */
float b=10;
clrscr();
printf("a=%d\n",a);
printf("b=%f\n",b);
printf("%d\n",5/2);
printf("%f\n",5.0/2);
printf("%f\n",5.0/2.0);
printf("%d\n", (int)5.0/2); /*explicit type casting */
printf("%f\n", (float)5/2);
printf("%f\n", (float)(5/2));
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

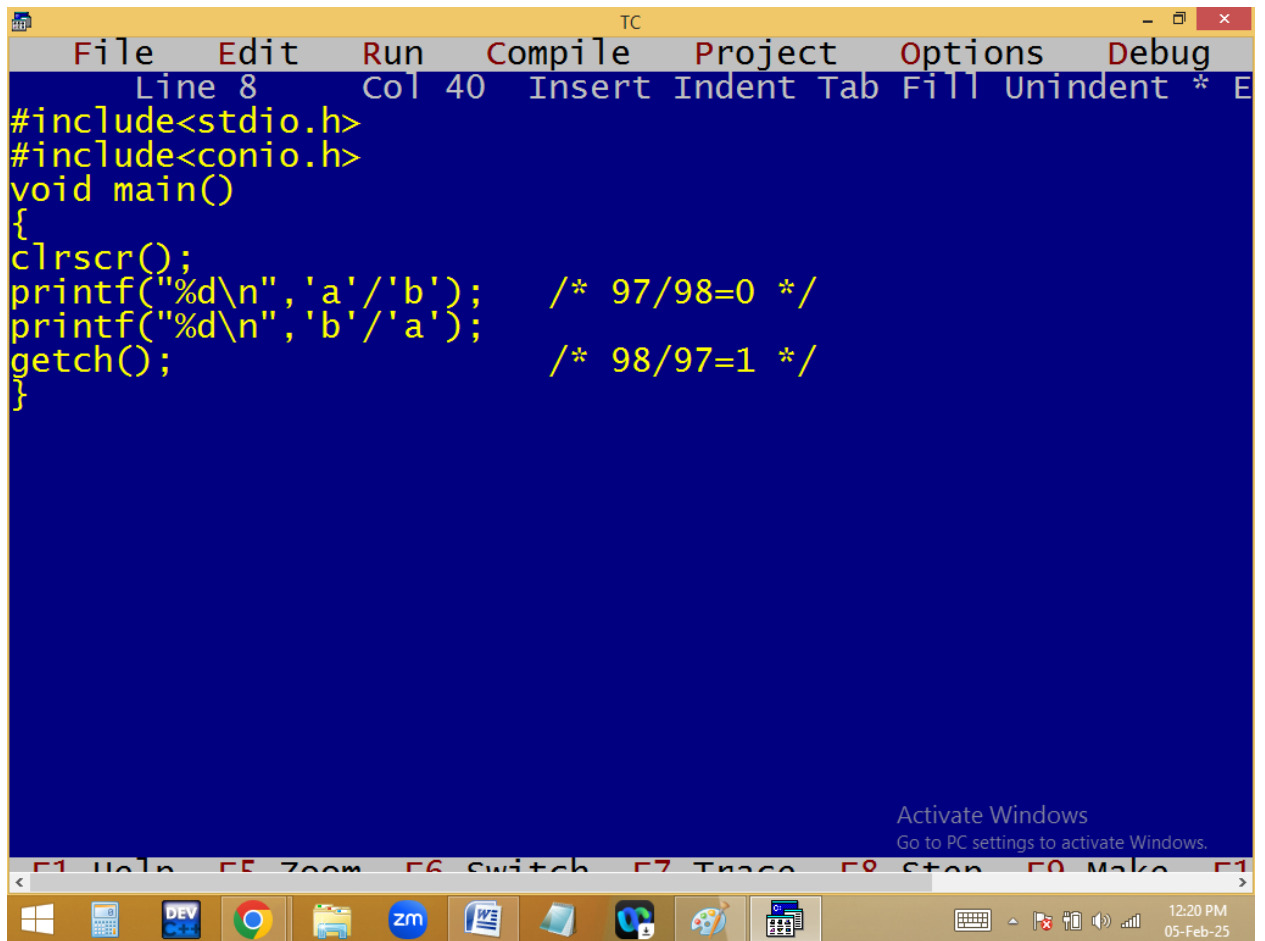
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar: 12:16 PM, 05-Feb-25

```
TC
a=10
b=10.000000
2
2.500000
2.500000
2
2.500000
2.000000
```

Activate Windows
Go to PC settings to activate Windows.

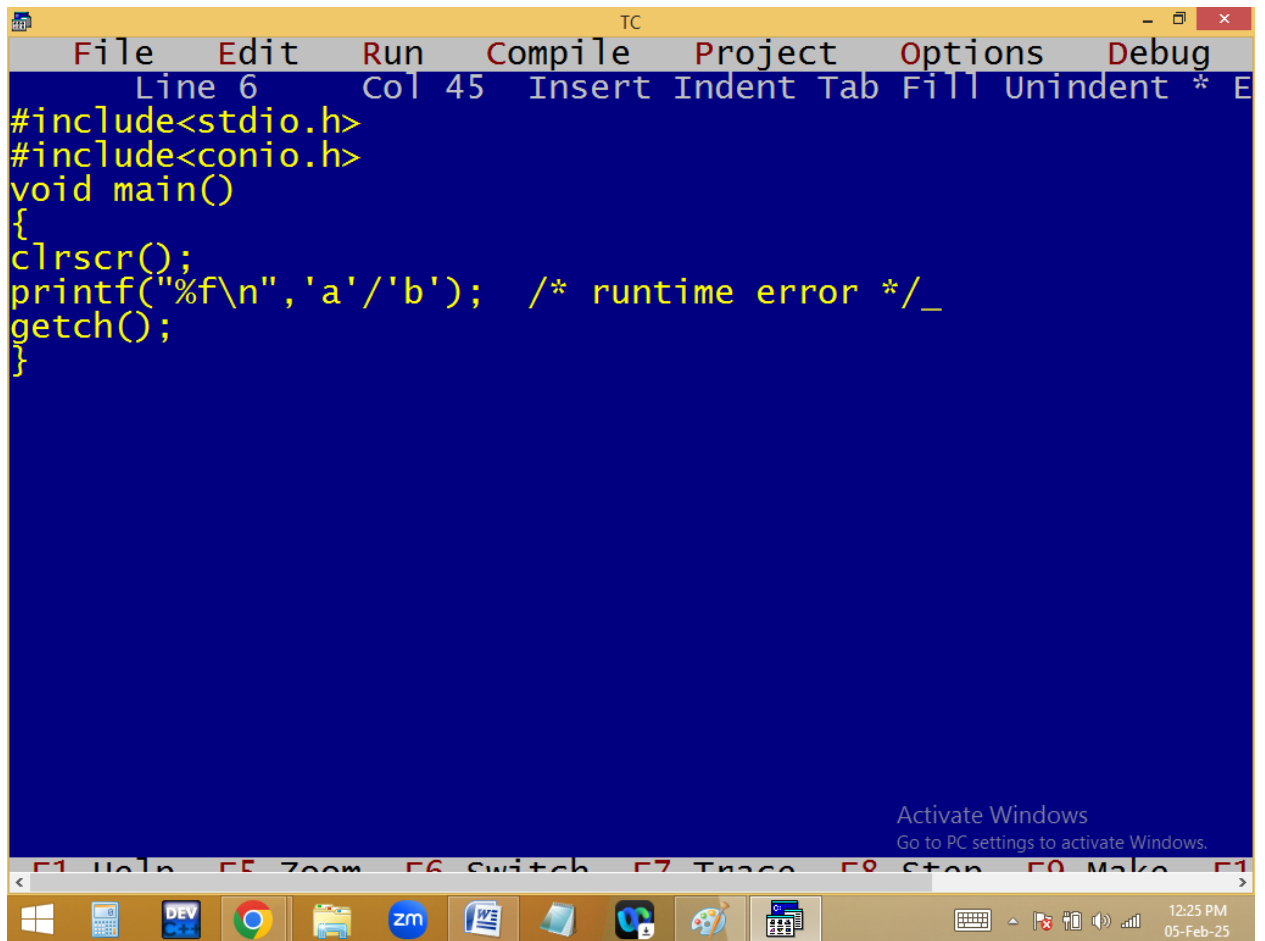
12:16 PM
05-Feb-25



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 40", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",'a'/'b');    /* 97/98=0 */
printf("%d\n",'b'/'a');    /* 98/97=1 */
getch();
}
```

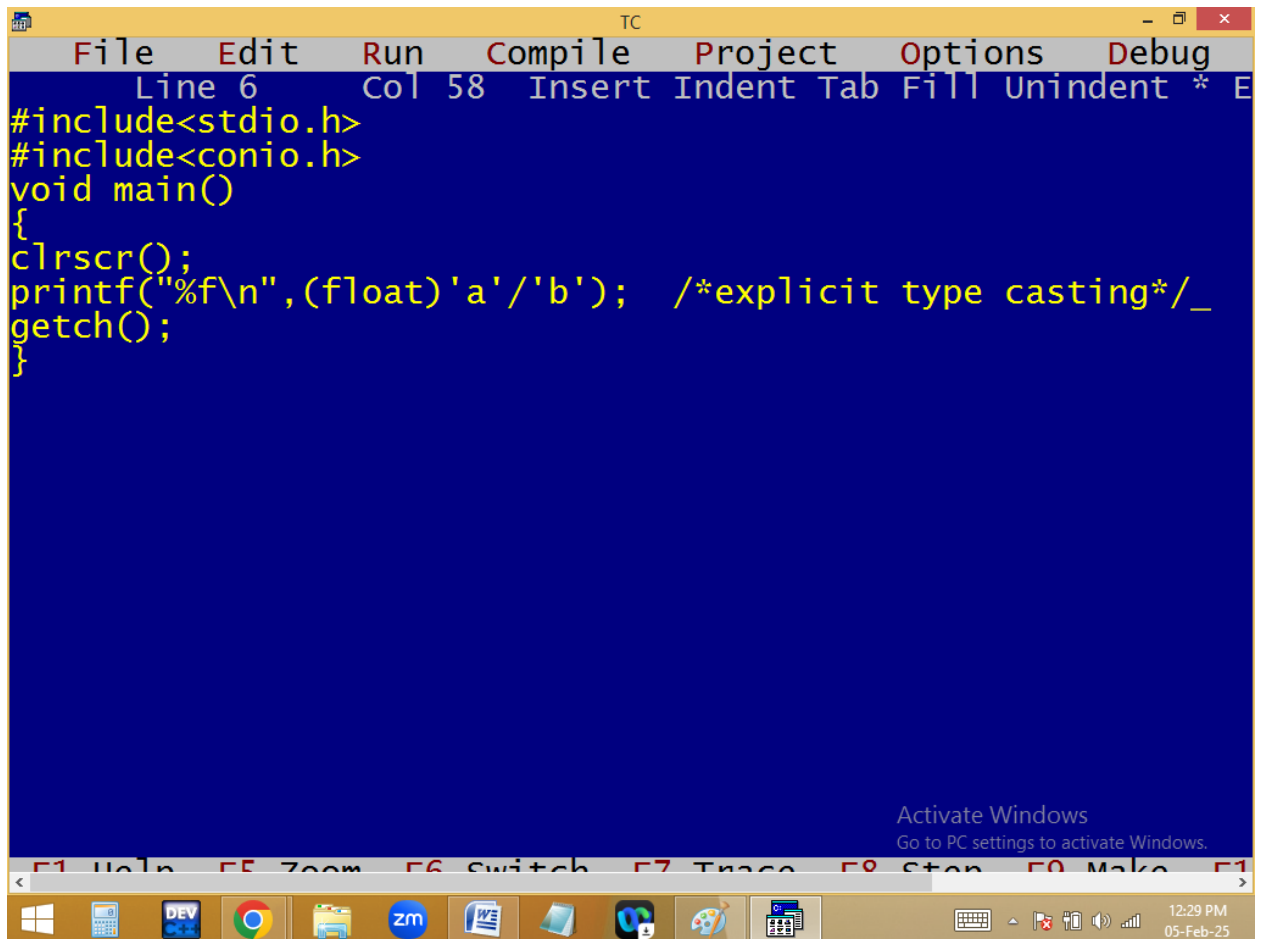
An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, with the text "Go to PC settings to activate Windows." Below the IDE window, the Windows taskbar is visible, showing icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and several other applications. The system tray on the right shows the date and time as "12:20 PM 05-Feb-25".



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 45", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",'a'/'b'); /* runtime error */_
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, a calculator, DEV C++, Google Chrome, a file explorer, Zoom, a document icon, a folder icon, a blue circular icon, a paint palette, and a calculator. On the right side of the taskbar, the system clock shows "12:25 PM" and "05-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, with the text "Go to PC settings to activate Windows."

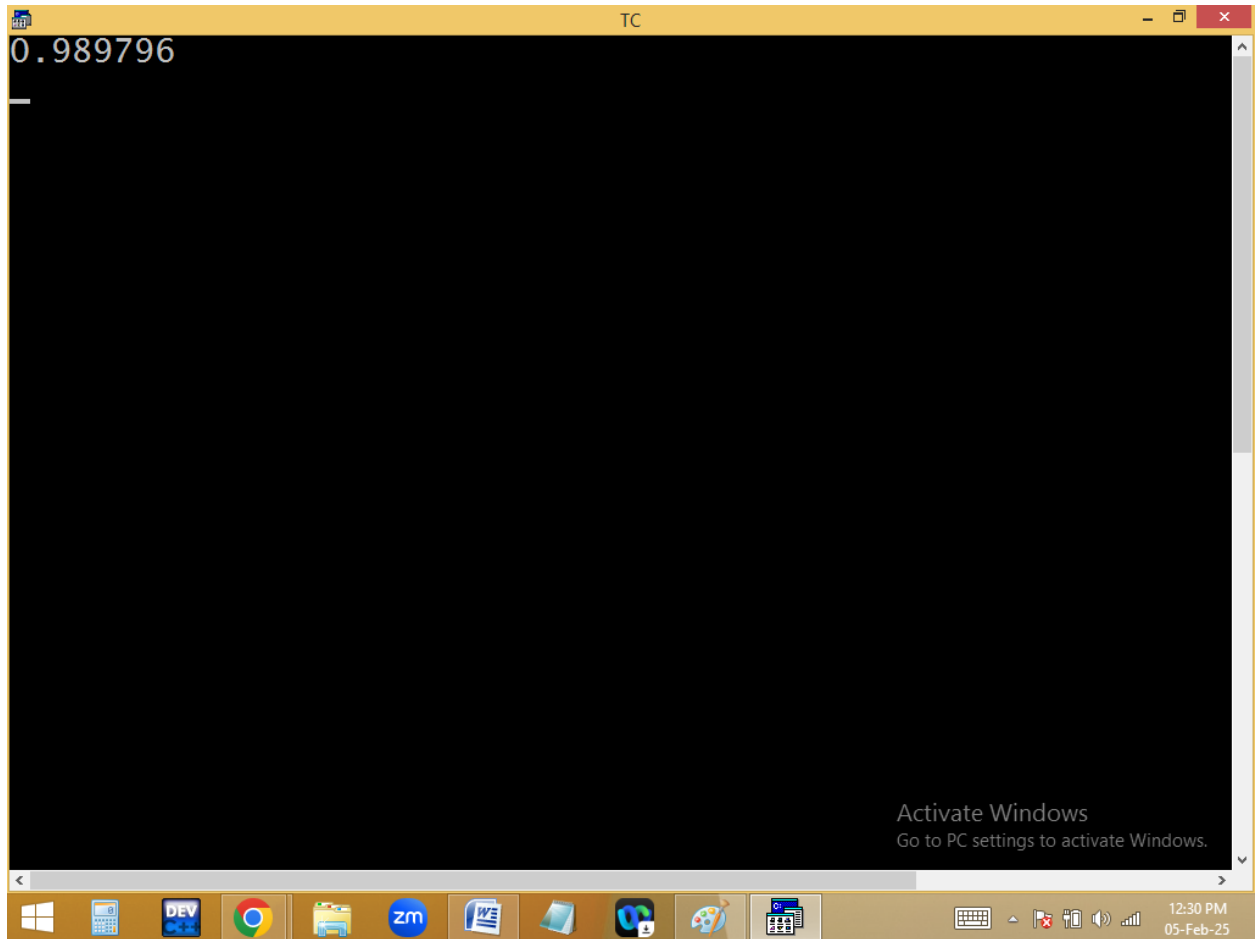


```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 58 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n", (float)'a'/'b'); /*explicit type casting*/
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:29 PM
05-Feb-25



$$123/10=12$$

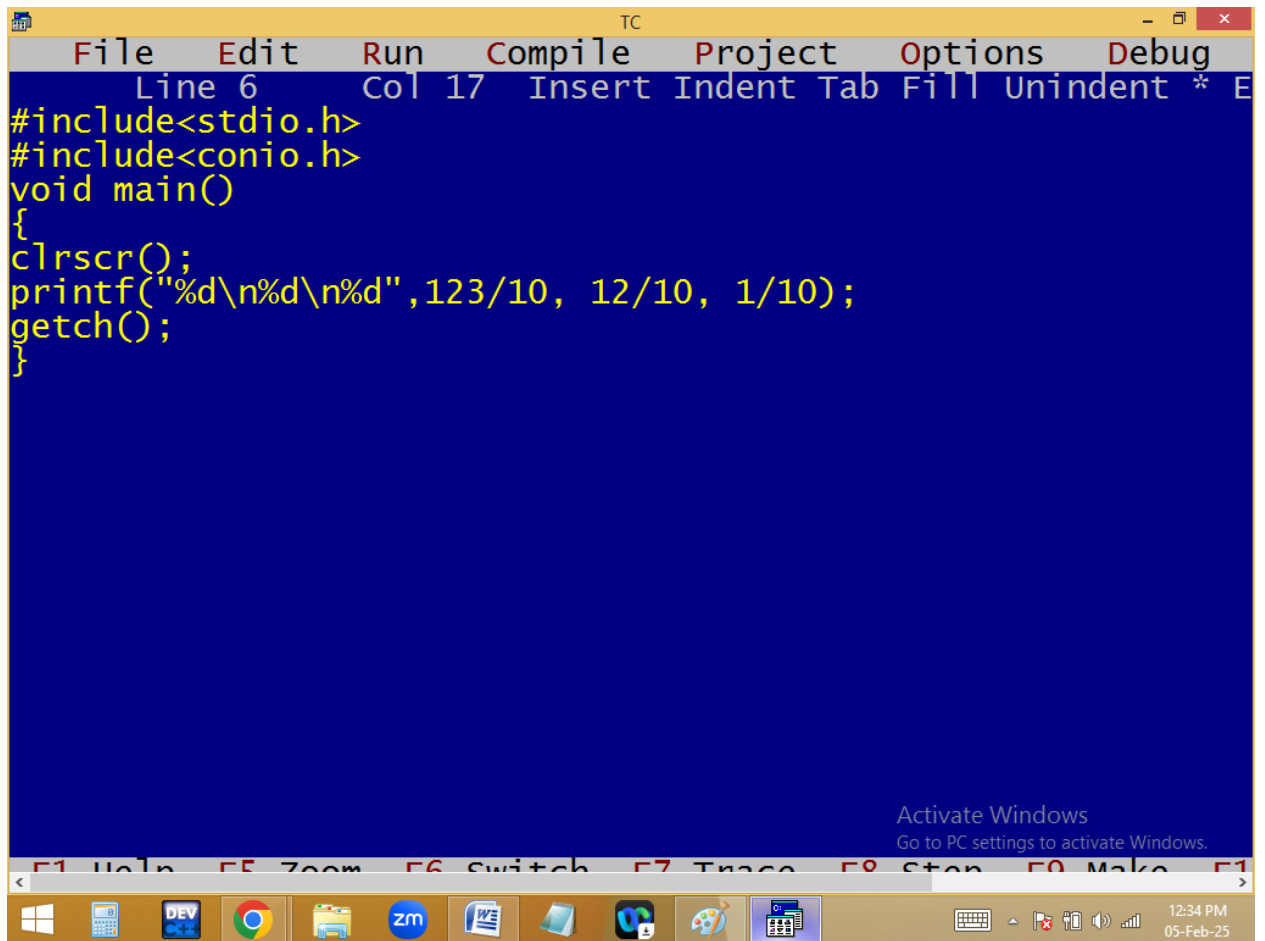
$$12/10=1$$

$$1/10=0$$

Note: Any no/10 removes the last digit.

$$123\%10=3$$

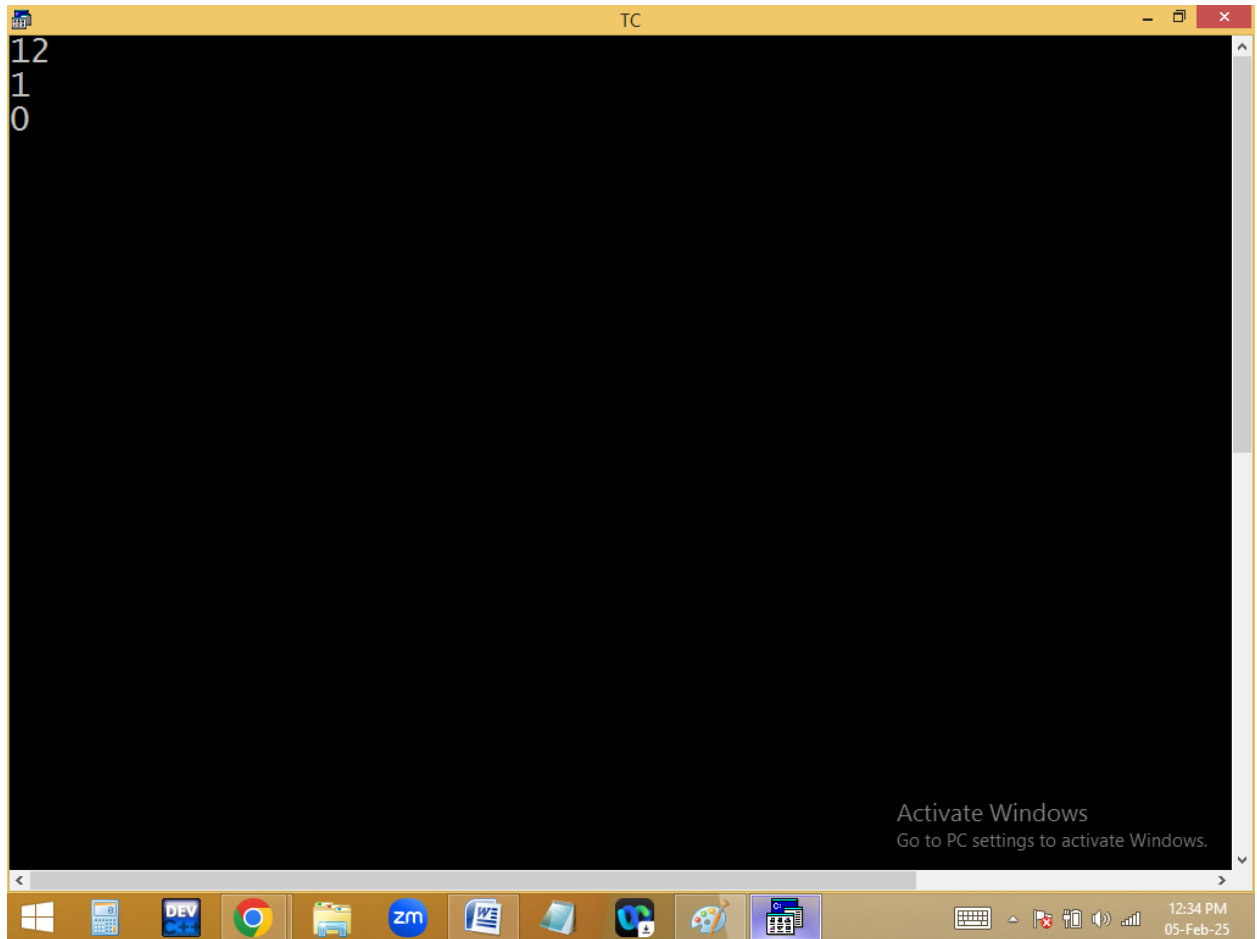
$$123/10=12$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 17", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d",123/10, 12/10, 1/10);
getch();
}
```

At the bottom of the window, there is a taskbar with icons for Windows, a calculator, DEV C++, Google Chrome, File Explorer, Zoom, a document icon, a folder icon, a paint palette, and a keyboard icon. The system tray on the right shows the time "12:34 PM" and the date "05-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

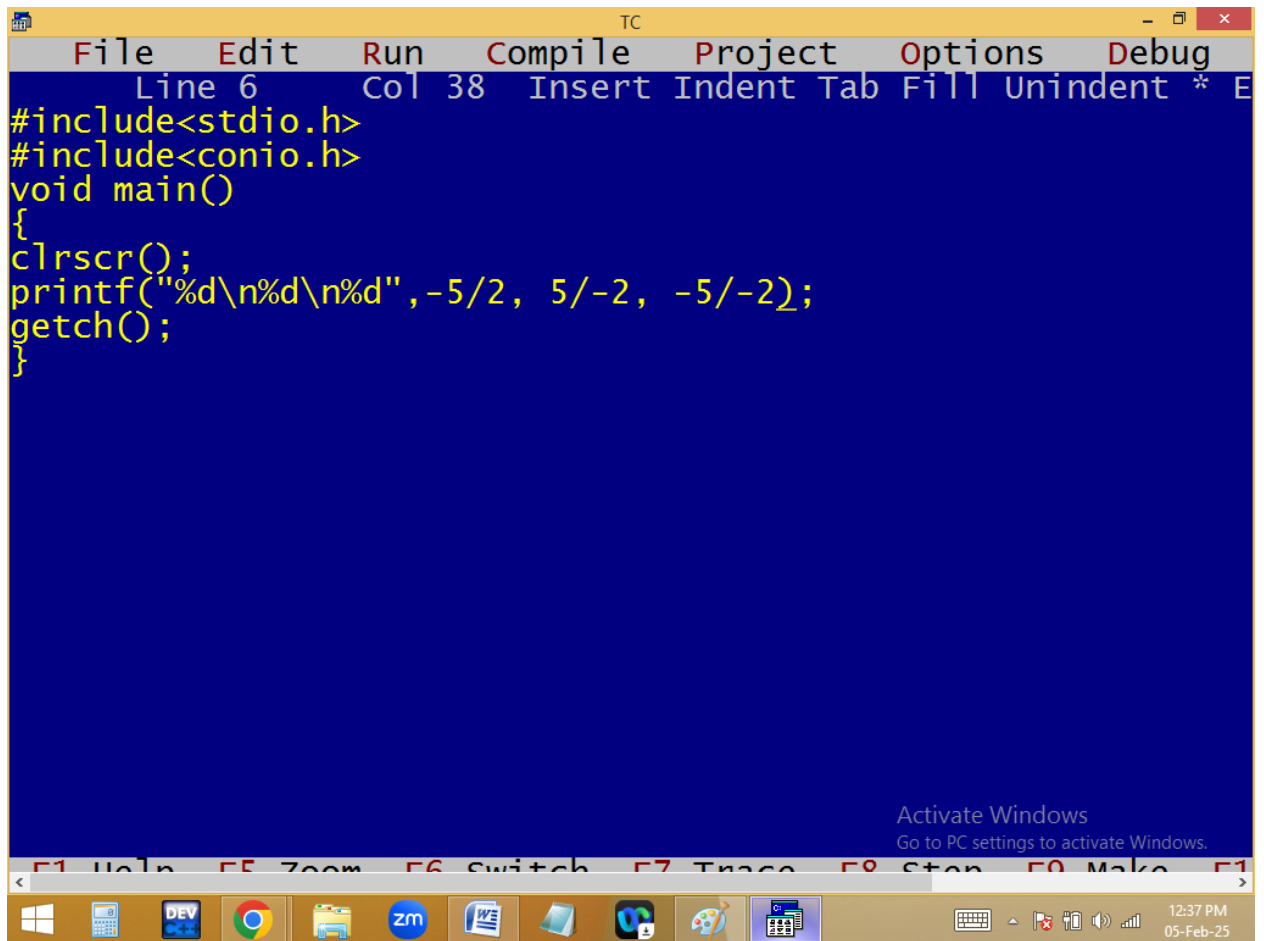


$$-5/2= -2$$

$$5/-2= -2$$

$$-5/-2= 2$$

Note: In division any one operand is negative then result also negative. If both are negative then result is positive.



TC

File Edit Run Compile Project Options Debug

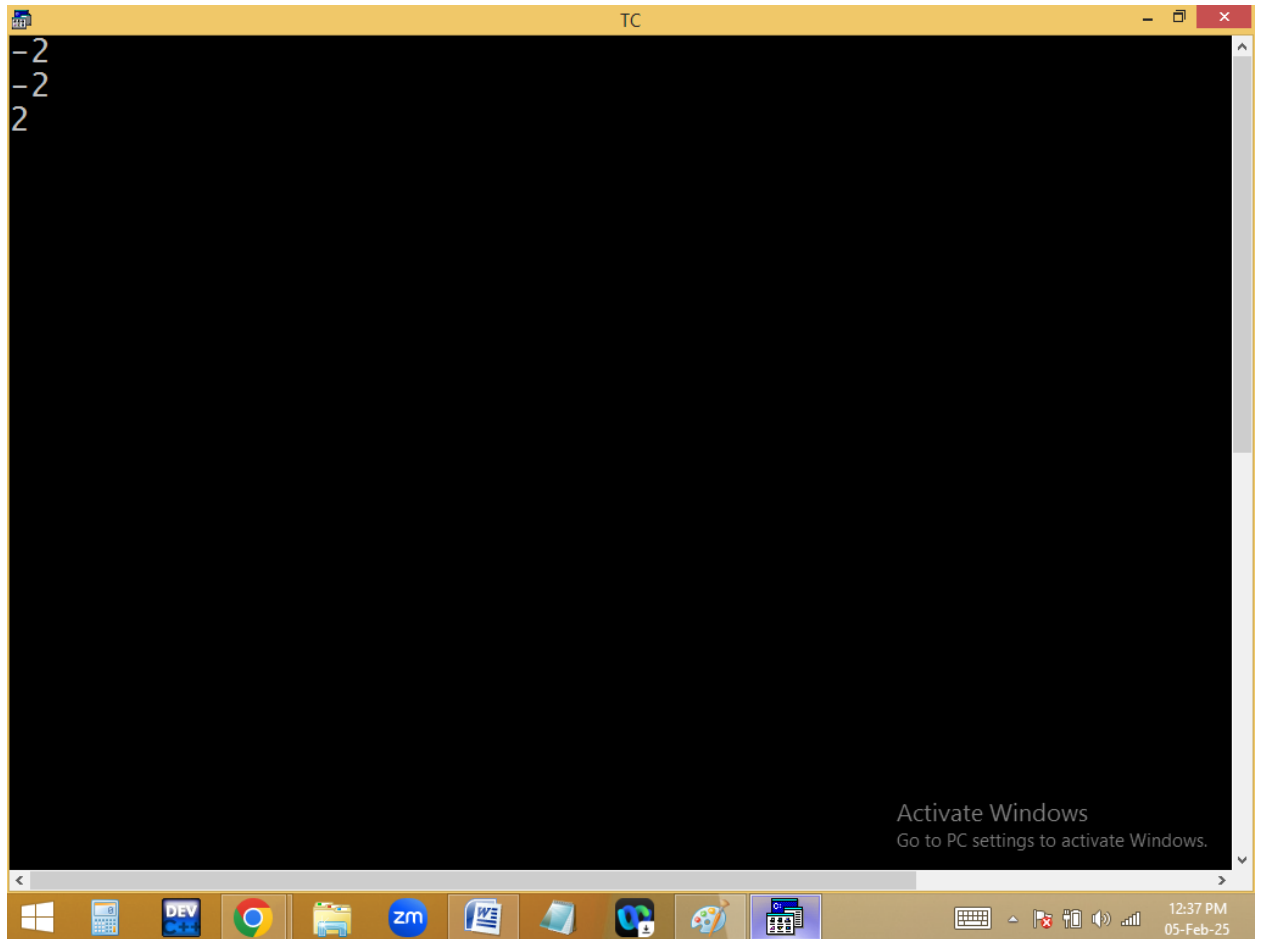
Line 6 Col 38 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d",-5/2, 5/-2, -5/-2);
getch();
}
```

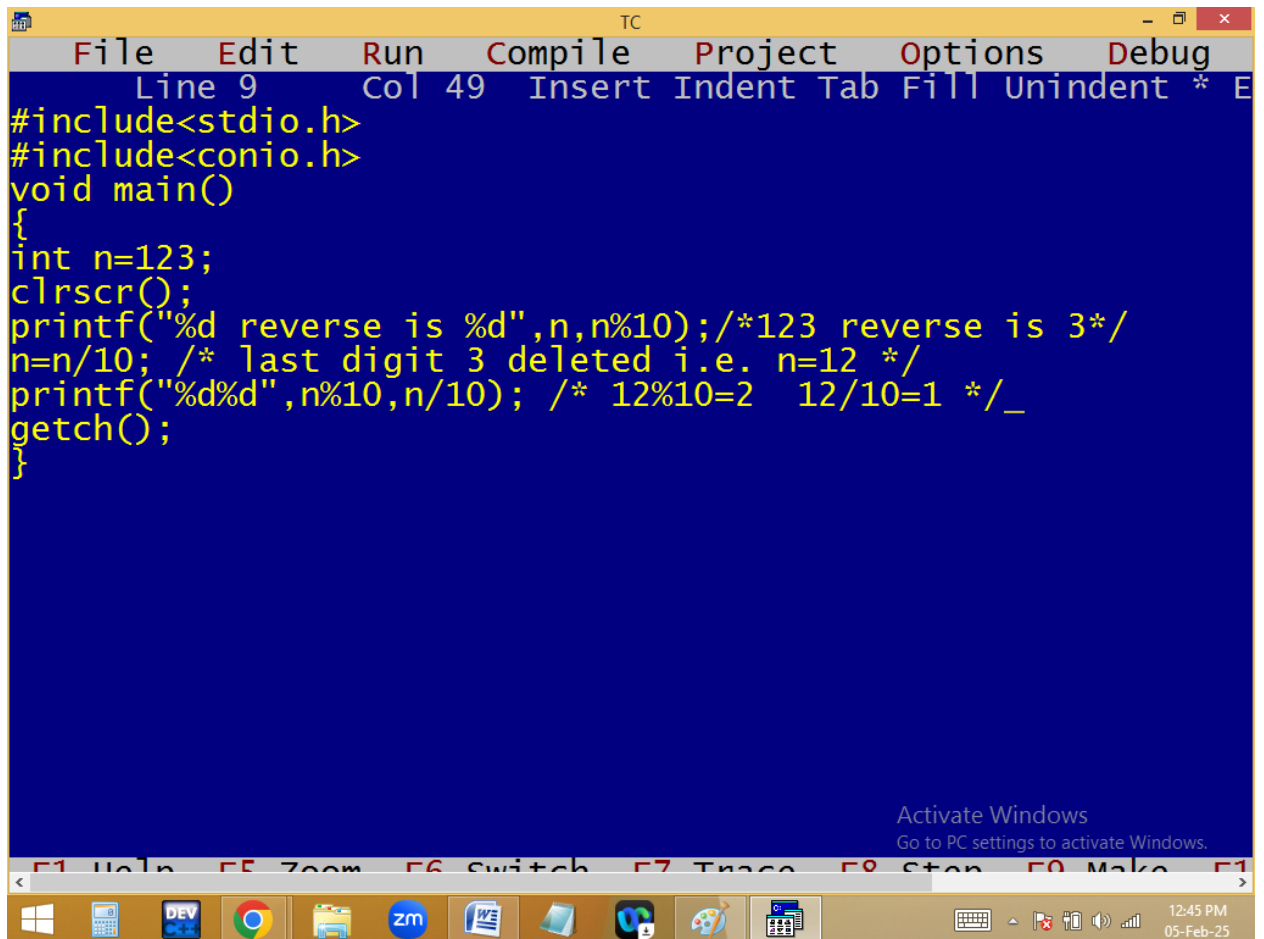
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:37 PM
05-Feb-25



Print 123 in reverse order without using loop?



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 9" and "Col 49". The code editor has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n=123;
clrscr();
printf("%d reverse is %d",n,n%10);/*123 reverse is 3*/
n=n/10; /* last digit 3 deleted i.e. n=12 */
printf("%d%d",n%10,n/10); /* 12%10=2  12/10=1 */_
getch();
}
```

At the bottom of the IDE window, there is a text box that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system tray on the right shows the time as 12:45 PM and the date as 05-Feb-25.

